

E215888_13

The equation $x^5 - xy + y^3 + 4 = 0$ has no integer solutions

Size $h = 48$, length $l = 12$

Theorem 1. *The equation*

$$x^5 - xy + y^3 + 4 = 0 \tag{1}$$

has no integer solutions.

The proof uses 43 auxiliary polynomials $H_0, \dots, H_{42}$, 8 of which are constants, and a graph on these vertices with 128 edges; to each edge $\{i, j\}$ we attach the Hilbert symbol $(H_i, H_j)_v$ of the values at a hypothetical solution. By the product formula, the product of these symbols over all edges and all places is 1. We show that the product over the edges is 1 at every odd prime outside a finite set S , compute it at the primes of S and at the real place, and obtain a contradiction. Apart from polynomial identities, which can be checked by expanding both sides, and finite checks of congruences, the proof uses only basic properties of the Legendre symbol and of the Hilbert symbol.

1 Legendre symbols and Hilbert symbols

For a prime p and a non-zero integer N , we write $v_p(N)$ for the exponent of p in the prime factorization of N . For an integer a and an odd prime p , the Legendre symbol (a/p) is 0 if $p \mid a$, 1 if a is a non-zero square modulo p , and -1 otherwise. It is multiplicative in a , and it depends only on a modulo p [1, Chapter 5].

For a prime p , let $\mathbb{Q}_p$ be the field of p -adic numbers, and put $\mathbb{Q}_\infty = \mathbb{R}$. For v a prime or $v = \infty$, and non-zero $a, b \in \mathbb{Q}$, the Hilbert symbol $(a, b)_v$ is 1 if the equation $z^2 = ar^2 + bs^2$ has a solution $(r, s, z) \neq (0, 0, 0)$ in $\mathbb{Q}_v$, and -1 otherwise. In particular, $(a, b)_v = 1$ for every v if this equation has a non-zero solution in $\mathbb{Q}$. For an odd integer t , we put $\varepsilon(t) = (t - 1)/2$ and $\omega(t) = (t^2 - 1)/8$. We use the following facts [2, Chapter III, Proposition 2 and Theorems 1 and 3].

Theorem 2. *Let a, a' and b be non-zero integers, and let v be a prime or $v = \infty$.*

- (i) $(a, b)_\infty = -1$ if and only if $a < 0$ and $b < 0$.
- (ii) Let p be an odd prime, and write $a = p^\alpha u$ and $b = p^\beta w$ with $p \nmid uw$. Then $(a, b)_p = (-1)^{\alpha\beta\varepsilon(p)}(u/p)^\beta(w/p)^\alpha$.
- (iii) Write $a = 2^\alpha u$ and $b = 2^\beta w$ with u and w odd. Then $(a, b)_2 = (-1)^{\varepsilon(u)\varepsilon(w) + \alpha\omega(w) + \beta\omega(u)}$.
- (iv) $(a, b)_v = 1$ for all but finitely many v , and $\prod_v (a, b)_v = 1$, the product over all primes v and $v = \infty$.
- (v) $(a, b)_v = (b, a)_v$ and $(aa', b)_v = (a, b)_v(a', b)_v$.

Lemma 3. *Let a and b be non-zero integers. If p is an odd prime with $p \nmid a$ and $(a/p) = 1$, then $(a, b)_p = 1$. If $a \equiv 1 \pmod{8}$, then $(a, b)_2 = 1$.*

Proof. The first claim is Theorem 2(ii) with $\alpha = 0$. For the second, $\varepsilon(a)$ and $\omega(a)$ are even, and Theorem 2(iii) with $\alpha = 0$ gives $(a, b)_2 = 1$. $\square$

Lemma 4. *Let p be a prime. For a non-zero integer X , write $X = p^j u$ with $p \nmid u$. If p is odd, put $c_p(X) = (j, e) \in \mathbb{F}_2^2$, where $(u/p) = (-1)^e$; if $p = 2$, put $c_2(X) = (j, e_1, e_2) \in \mathbb{F}_2^3$, where $u \equiv (-1)^{e_1} 5^{e_2} \pmod{8}$; all entries are taken modulo 2. Then $c_p(XX') = c_p(X) + c_p(X')$ and $(X, X')_p = (-1)^{\beta_p(c_p(X), c_p(X'))}$ for non-zero integers X, X' , where β_p is the symmetric bilinear form*

$$\begin{aligned} \beta_p((j, e), (j', e')) &= jj' \varepsilon(p) + je' + j'e && \text{if } p \text{ is odd,} \\ \beta_2((j, e_1, e_2), (j', e'_1, e'_2)) &= e_1 e'_1 + je'_2 + j'e_2. \end{aligned}$$

Proof. For odd p , this is Theorem 2(ii), as the Legendre symbol is multiplicative. For $p = 2$, the odd residues 1, 3, 5, 7 modulo 8 are $(-1)^{e_1} 5^{e_2}$ with $(e_1, e_2) = (0, 0), (1, 1), (0, 1), (1, 0)$, respectively; in each case $\varepsilon(u) \equiv e_1$ and $\omega(u) \equiv e_2 \pmod{2}$, so Theorem 2(iii) gives the formula, and c_2 is additive because $(e_1, e_2) \mapsto (-1)^{e_1} 5^{e_2}$ is an isomorphism from $\mathbb{F}_2^2$ to the group of odd residues modulo 8. $\square$

Lemma 5. *Let p be a prime, $k \geq 1$, and let X be a non-zero integer with $X \equiv r \pmod{p^k}$, where $0 \leq r < p^k$. Then $c_p(X) \in b + D$ for the vector b and the subspace D given as follows. If $r = 0$, then $b = 0$ and D is the whole space. Otherwise write $r = p^j r'$ with $p \nmid r'$, so that $j < k$. If p is odd, or if $p = 2$ and $k - j \geq 3$, then $b = c_p(r)$ and $D = 0$. If $p = 2$ and $k - j = 2$, then $b = (j, e_1, 0)$ with $r' \equiv (-1)^{e_1} \pmod{4}$, and D is spanned by $(0, 0, 1)$. If $p = 2$ and $k - j = 1$, then $b = (j, 0, 0)$, and D is spanned by $(0, 1, 0)$ and $(0, 0, 1)$.*

Proof. If $r \neq 0$, then $v_p(X) = j$ and $X/p^j \equiv r' \pmod{p^{k-j}}$. For odd p , the Legendre symbol of X/p^j is therefore that of r' . For $p = 2$, the residue of $X/2^j$ modulo 8 is that of r' if $k - j \geq 3$, and its residue modulo 4, which determines e_1 , is that of r' if $k - j = 2$. $\square$

Lemma 6. *Let p be a prime, and let E be a set of edges on a finite set of vertices. For every vertex i , let b_i be a vector and D_i a subspace of $\mathbb{F}_2^2$ (of $\mathbb{F}_2^3$ if $p = 2$), and put $\sigma_i = \sum_{j \sim i} b_j$, the sum over the edges $\{i, j\} \in E$. Suppose that $\beta_p(d, \sigma_i) = 0$ for every vertex i and every $d \in D_i$, and that $\beta_p(d, d') = 0$ for every edge $\{i, j\} \in E$ and all $d \in D_i$ and $d' \in D_j$. Then*

$$\sum_{\{i, j\} \in E} \beta_p(c_i, c_j) = \sum_{\{i, j\} \in E} \beta_p(b_i, b_j) \quad \text{whenever } c_i \in b_i + D_i \text{ for every } i.$$

By bilinearity, it suffices to check the hypotheses for d and d' in spanning sets of D_i and D_j .

Proof. Write $c_i = b_i + d_i$ with $d_i \in D_i$. As β_p is bilinear and symmetric, the left side equals the right side plus $\sum_i \beta_p(d_i, \sigma_i) + \sum_{\{i, j\} \in E} \beta_p(d_i, d_j)$, and both sums vanish. $\square$

Lemma 7. *Let $f(z) = az^3 + bz^2 + cz + d$ be a real polynomial with $a \neq 0$ and discriminant $\Delta < 0$. Then f has exactly one real root η , and $f(z)$ has the sign of $a(z - \eta)$ for every real $z \neq \eta$.*

Proof. The discriminant is $\Delta = a^4 \prod_{i < j} (\rho_i - \rho_j)^2$, where ρ_1, ρ_2, ρ_3 are the complex roots of f . If all roots were real, then $\Delta \geq 0$. Hence f has a non-real root ρ ; then $\bar{\rho} \neq \rho$ is also a root, and the third root η is real. Thus $f(z) = a(z - \eta)(z - \rho)(z - \bar{\rho}) = a(z - \eta)|z - \rho|^2$ for real z , and $|z - \rho|^2 > 0$. $\square$

2 The graph

Put $F(x, y) = x^5 - xy + y^3 + 4$, so that (1) is the equation $F(x, y) = 0$. The vertices are the polynomials

$$\begin{aligned} H_0 &= x^2 - x - y, \\ H_1 &= x^2 + x - y + 1, \\ H_2 &= -y, \\ H_3 &= -x - y, \\ H_4 &= -y - 1, \\ H_5 &= -y + 1, \\ H_6 &= x - y, \\ H_7 &= -2x - y, \\ H_8 &= -x - y - 1, \\ H_9 &= -x - y + 1, \\ H_{10} &= -y - 2, \\ H_{11} &= -y + 2, \\ H_{12} &= 2x - y, \\ H_{13} &= -2x - y - 1, \\ H_{14} &= -x - y - 2, \\ H_{15} &= -x - y + 2, \\ H_{16} &= x - y - 2, \\ H_{17} &= -2x - y - 2, \\ H_{18} &= -2x - y + 2, \\ H_{19} &= x^2 + y, \\ H_{20} &= x^2 - x + y, \\ H_{21} &= x^2 + y + 1, \\ H_{22} &= x^2 + x + y, \\ H_{23} &= x^2 - y - 1, \\ H_{24} &= x^2 + x - y, \\ H_{25} &= x^2 - x + y + 1, \\ H_{26} &= x^2 + y - 2, \\ H_{27} &= x^2 + y + 2, \\ H_{28} &= x^2 + x + y + 1, \\ H_{29} &= x^2 - x - y + 1, \\ H_{30} &= x^2 - y - 2, \\ H_{31} &= x^2 - y + 2, \\ H_{32} &= x^2 + x - y - 1, \\ H_{33} &= x^2 - x - 2y, \\ H_{34} &= x^2 + x + 2y, \\ H_{35} &= 1, \\ H_{36} &= 10, \\ H_{37} &= 15, \end{aligned}$$

$$\begin{aligned}
H_{38} &= -1, \\
H_{39} &= -5, \\
H_{40} &= -6, \\
H_{41} &= 5, \\
H_{42} &= 2.
\end{aligned}$$

The edges are $\{0, 2\}, \{0, 5\}, \{0, 7\}, \{0, 8\}, \{0, 11\}, \{0, 13\}, \{0, 20\}, \{0, 31\}, \{1, 4\}, \{1, 11\}, \{1, 30\}, \{2, 3\}, \{2, 18\}, \{2, 20\}, \{3, 4\}, \{3, 6\}, \{3, 12\}, \{3, 13\}, \{3, 16\}, \{3, 17\}, \{3, 21\}, \{3, 23\}, \{3, 24\}, \{3, 33\}, \{3, 34\}, \{4, 8\}, \{4, 14\}, \{4, 16\}, \{4, 22\}, \{4, 24\}, \{4, 28\}, \{4, 33\}, \{5, 8\}, \{5, 9\}, \{5, 16\}, \{5, 17\}, \{5, 20\}, \{5, 24\}, \{5, 26\}, \{5, 29\}, \{5, 32\}, \{5, 33\}, \{5, 34\}, \{6, 15\}, \{6, 21\}, \{6, 23\}, \{6, 27\}, \{6, 30\}, \{6, 34\}, \{7, 11\}, \{7, 15\}, \{7, 22\}, \{7, 25\}, \{7, 27\}, \{8, 12\}, \{8, 16\}, \{8, 19\}, \{8, 26\}, \{8, 29\}, \{8, 33\}, \{8, 34\}, \{9, 21\}, \{9, 23\}, \{10, 16\}, \{10, 18\}, \{11, 15\}, \{11, 17\}, \{11, 22\}, \{11, 23\}, \{12, 26\}, \{12, 31\}, \{13, 15\}, \{13, 16\}, \{13, 20\}, \{13, 22\}, \{13, 25\}, \{13, 33\}, \{14, 19\}, \{14, 22\}, \{14, 24\}, \{14, 33\}, \{15, 16\}, \{15, 22\}, \{15, 25\}, \{15, 26\}, \{15, 33\}, \{16, 19\}, \{16, 20\}, \{16, 30\}, \{16, 34\}, \{17, 22\}, \{17, 23\}, \{17, 30\}, \{17, 32\}, \{17, 33\}, \{18, 23\}, \{18, 31\}, \{18, 34\}, \{19, 26\}, \{19, 27\}, \{19, 30\}, \{19, 31\}, \{22, 24\}, \{22, 26\}, \{22, 30\}, \{22, 33\}, \{24, 31\}, \{25, 27\}, \{25, 32\}, \{26, 30\}, \{26, 33\}, \{27, 31\}, \{28, 31\}, \{29, 33\}, \{30, 32\}, \{2, 36\}, \{3, 37\}, \{5, 38\}, \{6, 39\}, \{8, 38\}, \{9, 39\}, \{12, 40\}, \{20, 38\}, \{21, 41\}, \{22, 38\}, \{28, 42\}, \{29, 41\}, \{34, 41\}. For a vertex i , we write $j \sim i$ if $\{i, j\}$ is an edge. For every non-constant vertex i , Table 1 gives a positive integer m_i and polynomials W_i , A_i and B_i with integer coefficients such that$

$$m_i^2 \prod_{j \sim i} H_j - W_i^2 = A_i F + B_i H_i. \quad (2)$$

For every edge $\{i, j\}$ between two non-constant vertices, Table 2 gives a positive integer k_{ij} and polynomials α_{ij} , β_{ij} and γ_{ij} with integer coefficients such that

$$\alpha_{ij} F + \beta_{ij} H_i + \gamma_{ij} H_j = k_{ij}. \quad (3)$$

These identities can be checked by expanding both sides. For the signs of the vertices, we regard F as a polynomial in y ; it has degree 3 and leading coefficient 1, and its discriminant is

$$\Delta(x) = -27x^{10} - 216x^5 + 4x^3 - 432.$$

For $i \in I = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34\}$, the vertex H_i has the form $k_i y + r_i(x)$ with a non-zero integer k_i , and Table 3 lists the polynomial $\varphi_i(x) = k_i^3 F(x, -r_i(x)/k_i)$, which has integer coefficients.

3 Proof of Theorem 1

Suppose that $(x, y) \in \mathbb{Z}^2$ is a solution of (1), so that $F(x, y) = 0$; from now on the polynomials denote their values at (x, y) .

Step 1: the values are non-zero. The non-constant H_i are non-zero by Step 2.

Step 2: signs. *The signs of H_i for $i \in I$.* If $\Delta(x) < 0$, then by Lemma 7, applied to $F(x, z)$ as a polynomial in z , y is its only real root, and $F(x, z)$ has the sign of $(z - y)$ for $z \neq y$. For $i \in I$ and $z = -r_i(x)/k_i$, we have $z - y = -H_i/k_i$, so if moreover $\varphi_i(x) = k_i^3 F(x, z) \neq 0$, then $H_i \neq 0$, and H_i has the sign of $-\varphi_i(x)$. Expanding $\Delta(5 + t)$ and the $\varphi_i(5 + t)$ as polynomials in t gives in each case coefficients of one sign and a non-zero constant term, and likewise for $\Delta(-5 - t)$ and the $\varphi_i(-5 - t)$. Hence, for $x \geq 5$ and for $x \leq -5$, we have $\Delta(x) < 0$, and $\varphi_i(x)$ is non-zero and has the sign of the constant term of the corresponding expansion; the resulting

signs of the H_i are listed in Table 3. It remains to consider $-4 \leq x \leq 4$. Checking the residues of y , we find that for $x \in \{-3, -1, 1, 3\}$, we have $F(x, y) \not\equiv 0 \pmod{2}$ for every y , for $x = -2$ and $x = 4$, we have $F(x, y) \not\equiv 0 \pmod{3}$ for every y , for $x = 0$, we have $F(x, y) \not\equiv 0 \pmod{7}$ for every y , for $x = 2$, we have $F(x, y) \not\equiv 0 \pmod{11}$ for every y , and for $x = -4$, we have $F(x, y) \not\equiv 0 \pmod{19}$ for every y . Hence (1) has no solution with $-4 \leq x \leq 4$.

For a place v , put $B_v = \prod_{\{i,j\}} (H_i, H_j)_v$, the product over the edges. By Theorem 2(i), $(H_i, H_j)_\infty = -1$ exactly when $H_i < 0$ and $H_j < 0$. Here $H_i > 0$ for $i \in \{35, 36, 37, 41, 42\}$ and $H_i < 0$ for $i \in \{38, 39, 40\}$. The other signs are given in Table 3 for $x \geq 5$ and for $x \leq -5$. In both cases, counting the edges whose two values are negative gives $B_\infty = 1$.

Step 3: odd primes outside S . Let $S = \{2, 3, 5, 7, 11, 13, 17\}$ be the set of primes dividing 2, the m_i , the k_{ij} and the constant vertices. Let $p \notin S$ be an odd prime; we show that $B_p = 1$. For an edge $\{i, j\}$ such that p divides neither H_i nor H_j , $(H_i, H_j)_p = 1$ by Theorem 2(ii). Now let i be a vertex with $p \mid H_i$. Then H_i is not constant, as $p \notin S$. For every edge $\{i, j\}$, $p \nmid H_j$: if H_j is constant, because $p \notin S$, and otherwise by (3), because $p \nmid k_{ij}$. In particular, no edge has both values divisible by p . By Theorem 2(v), the edges at i contribute $\prod_{j \sim i} (H_i, H_j)_p = (H_i, \Pi_i)_p$, where $\Pi_i = \prod_{j \sim i} H_j$ is not divisible by p . By (2), $m_i^2 \Pi_i \equiv W_i^2 \pmod{p}$, and $p \nmid m_i$, so Π_i is a non-zero square modulo p , and $(H_i, \Pi_i)_p = (\Pi_i, H_i)_p = 1$ by Lemma 3. Hence $B_p = 1$.

Step 4: the primes in S . Let $p \in S$. By Lemma 4, $B_p = (-1)^N$, where $N = \sum_{\{i,j\}} \beta_p(c_p(H_i), c_p(H_j))$, the sum over the edges. Suppose that $x \equiv a$ and $y \equiv b \pmod{p^k}$. Then $H_i \equiv H_i(a, b) \pmod{p^k}$, and Lemma 5 gives a vector b_i and a subspace D_i with $c_p(H_i) \in b_i + D_i$; for a constant vertex, we take $b_i = c_p(H_i)$ and $D_i = 0$. If the hypotheses of Lemma 6 hold for the edges, then $N = \sum_{\{i,j\}} \beta_p(b_i, b_j)$, so B_p is determined by a, b and k . Starting from the solutions of (1) modulo p , we refine a residue class $x \equiv a, y \equiv b \pmod{p^k}$ to the classes modulo p^{k+1} that contain solutions of (1) modulo p^{k+1} , until these hypotheses hold, or until no solution remains. Table 4 lists the resulting terminal classes together with the value of B_p ; every integer solution lies in one of them. The result is $B_2 = -1, B_3 = 1, B_5 = 1, B_7 = 1, B_{11} = 1, B_{13} = 1, B_{17} = 1$.

Step 5: contradiction. By Theorem 2(iv), applied to each edge, $\prod_v B_v = 1$, the product over all primes and $v = \infty$. By Step 3, the factors with $p \notin S$ are 1, so $\prod_{p \in S} B_p = B_\infty = 1$ by Step 2. But by Step 4, $\prod_{p \in S} B_p = -1$. This contradiction proves the theorem. $\square$

References

- [1] K. Ireland and M. Rosen, *A Classical Introduction to Modern Number Theory*, 2nd ed., Graduate Texts in Mathematics 84, Springer, New York, 1990.
- [2] J.-P. Serre, *A Course in Arithmetic*, Graduate Texts in Mathematics 7, Springer, New York, 1973.

Table 1: The data of the identities (2).

$i = 0, m = 4$, neighbours 2, 5, 7, 8, 11, 13, 20, 31	
W	$-4x^5 + 16x^4 - 56x^3 - 4x^2 + 48x + 80$
A	$48x^8y - 376x^8 + 1044x^5 - 144x^4y^3 + 936x^4y^2 + 1552x^4y + 1698x^4$ $-704x^3 + 64x^2y^3 - 192x^2y^2 + 128x^2y + 1760x^2 + 128xy^4 - 352xy^3$ $-3020xy^2 - 4144xy - 3102x + 96y^5 - 576y^4 - 568y^3 - 4738y^2 - 1182y - 1600$
B	$-48x^{11}y + 376x^{11} - 48x^{10}y + 376x^{10} - 48x^9y^2 + 328x^9y + 376x^9$ $-96x^8y^2 + 704x^8y - 684x^8 + 96x^7y^3 - 656x^7y^2 - 848x^7y - 2254x^7$ $-48x^6y^4 + 376x^6y^3 + 48x^6y^2 - 1724x^6y - 750x^6 + 48x^5y^4$ $-280x^5y^3 - 800x^5y^2 - 3978x^5y - 750x^5 - 48x^4y^5 + 424x^4y^4$ $-232x^4y^3 + 656x^4y^2 + 524x^4y - 272x^4 - 16x^3y^5 + 320x^3y^4$ $-300x^3y^3 + 2744x^3y^2 + 2782x^3y - 4192x^3 + 112x^2y^6 - 624x^2y^5$ $-1064x^2y^4 - 1502x^2y^3 - 124x^2y^2 - 4338x^2y - 8184x^2 + 80xy^6$ $-64xy^5 - 1732xy^4 + 1482xy^3 + 3522xy^2 - 7026xy + 3976x + 112y^7 - 624y^6$ $-584y^5 - 4242y^4 - 3486y^3 - 3936y^2 - 18952y - 4728$
$i = 1, m = 4$, neighbours 4, 11, 30	
W	$2x^5 + 6x^4 + 8x^3 + 8x^2 - 6$
A	$-2x^5 - 43x^4 - 68x^3 - 128x^2 + 2xy^2 - 16xy - 87x + 35y^2 + 23y - 46$
B	$-2x^8 + 21x^7 - 2x^6y - 19x^6 + 23x^5y - 2x^5 - 4x^4y^2 - 26x^4y - 52x^4$ $+2x^3y^3 - 8x^3y^2 - 65x^3y + 4x^3 + 37x^2y^3 - 14x^2y^2 - 29x^2y$ $+228x^2 + 2xy^4 + 21xy^3 - 43xy^2 - 30xy + 136x + 35y^4 + 58y^3 + 28y^2 + 184y + 212$
$i = 2, m = 1$, neighbours 36, 0, 3, 18, 20	
W	$x^4 - 4x^3 + 6x^2 - 4x - 4$
A	$-x^3 + 8x^2 - 8x + 30y - 4$
B	$-10x^4y + 81x^4 - x^3y^2 + 20x^3y - 78x^3 + 8x^2y^2 + 10x^2y + 28x^2$ $+22xy^2 - 50xy + 4x + 40y^3 - 24y^2 + 120$
$i = 3, m = 1$, neighbours 37, 2, 4, 6, 12, 13, 16, 17, 21, 23, 24, 33, 34	
W	$30x^4 + 90x^3 - 120x$
A	$120x^{10}y^2 + 120x^{10}y - 180x^9y^3 - 120x^9y^2 + 60x^9y - 30x^8y^4$ $-570x^8y^3 - 1020x^8y^2 - 480x^8y + 105x^7y^5 + 570x^7y^4$ $+195x^7y^3 - 750x^7y^2 - 480x^7y + 195x^6y^5 + 1110x^6y^4$ $+1785x^6y^3 + 990x^6y^2 + 120x^6y - 15x^5y^7 - 195x^5y^6 + 240x^5y^5$ $+1935x^5y^4 + 3345x^5y^3 + 1710x^5y^2 - 120x^5y - 15x^4y^7$ $-240x^4y^6 + 60x^4y^5 + 1650x^4y^4 + 4095x^4y^3 + 3090x^4y^2$ $+360x^4y + 15x^3y^8 - 390x^3y^7 - 2775x^3y^6 - 4935x^3y^5$ $-3180x^3y^4 + 2475x^3y^3 + 5490x^3y^2 + 2400x^3y - 900x^3 - 45x^2y^8$ $-750x^2y^7 - 3090x^2y^6 - 6855x^2y^5 - 9195x^2y^4 - 4995x^2y^3$ $+1410x^2y^2 + 1680x^2y - 5400x^2 + 75xy^9 + 840xy^8 + 1650xy^7 - 60xy^6$ $-5490xy^5 - 11610xy^4 - 13215xy^3 - 7050xy^2 - 900xy - 8100x + 15y^{10} + 255y^9$ $+570y^8 + 570y^7 - 450y^6 - 2385y^5 - 5925y^4 - 9630y^3 + 900y^2 - 8100y$
B	$45x^4y^9 + 105x^4y^8 + 30x^4y^7 - 1185x^4y^6 - 1260x^4y^5$ $+1410x^4y^4 + 4305x^4y^3 + 8730x^4y^2 - 8460x^4y - 7200x^4 + 15x^3y^{10}$ $-285x^3y^9 - 2085x^3y^8 - 4470x^3y^7 - 7035x^3y^6 - 8355x^3y^5$ $-5085x^3y^4 + 1680x^3y^3 + 18600x^3y^2 + 9540x^3y - 21600x^3$ $-45x^2y^{10} - 510x^2y^9 - 1035x^2y^8 - 1155x^2y^7 - 4785x^2y^6$ $-9405x^2y^5 - 10635x^2y^4 - 11490x^2y^3 + 10740x^2y^2 + 36600x^2y$ $-3600x^2 + 60xy^{11} + 630xy^{10} + 1545xy^9 + 2160xy^8 + 3330xy^7 + 2925xy^6$ $-3225xy^5 - 10695xy^4 - 29070xy^3 - 30060xy^2 + 18420xy - 7200x + 15y^{12} + 195y^{11}$ $+90y^{10} - 930y^9 - 2070y^8 - 2565y^7 - 4845y^6 - 11670y^5 - 8640y^4 - 31800y^3$ $-38520y^2 + 3600y - 32400$
$i = 4, m = 1$, neighbours 1, 3, 8, 14, 16, 22, 24, 28, 33	
W	$-3x^4 - 3x^3 + 3x + 3$
A	$-x^9 - 2x^8y - 4x^8 - x^7y - 3x^7 + 2x^6y^3 + 13x^6y^2 + 23x^6y + 10x^6$ $+x^5y^4 + 11x^5y^3 + 47x^5y^2 + 63x^5y + 28x^5 - x^4y^4 + 10x^4y^3$ $+46x^4y^2 + 62x^4y + 34x^4 - 2x^3y^5 - 20x^3y^4 - 66x^3y^3$ $-69x^3y^2 - 13x^3y + 14x^3 - 12x^2y^5 - 93x^2y^4 - 228x^2y^3$ $-249x^2y^2 - 128x^2y - 26x^2 - 4xy^6 - 42xy^5 - 173xy^4 - 333xy^3 - 388xy^2$ $-261xy - 76x - y^7 - 5y^6 - 44y^5 - 153y^4 - 324y^3 - 471y^2 - 367y - 110$
B	$-5x^4y^6 - 19x^4y^5 - 14x^4y^4 + 86x^4y^3 + 191x^4y^2 + 194x^4y$ $+104x^4 - 2x^3y^7 - 28x^3y^6 - 97x^3y^5 - 38x^3y^4 + 57x^3y^3$ $-40x^3y^2 - 52x^3y + 38x^3 - 13x^2y^7 - 87x^2y^6 - 141x^2y^5$

$$\begin{aligned}
& -72x^2y^4 - 184x^2y^3 - 390x^2y^2 - 341x^2y - 95x^2 - 4xy^8 - 32xy^7 \\
& -117xy^6 - 181xy^5 - 265xy^4 - 412xy^3 - 537xy^2 - 648xy - 286x - y^9 - 2y^8 \\
& -32y^7 - 111y^6 - 235y^5 - 428y^4 - 559y^3 - 847y^2 - 1037y - 431
\end{aligned}$$

$i = 5, m = 1$, neighbours 38, 0, 8, 9, 16, 17, 20, 24, 26, 29, 32, 33, 34

W $-10x^4 + 25x^3 + 40x^2 - 60x - 45$

A $2x^{15} + 3x^{14}y - 4x^{14} - x^{13}y^2 - 15x^{13}y - 14x^{13} - 3x^{12}y^3$
 $-15x^{12}y^2 - 12x^{12}y + 32x^{12} - x^{11}y^4 + x^{11}y^3 + 15x^{11}y^2$
 $+112x^{11}y + 30x^{11} + 7x^{10}y^4 + 23x^{10}y^3 + 137x^{10}y^2 - 27x^{10}y$
 $-108x^{10} + 2x^9y^5 + 18x^9y^4 + 100x^9y^3 - 144x^9y^2 - 339x^9y$
 $+16x^9 + 17x^8y^5 + 49x^8y^4 - 241x^8y^3 - 435x^8y^2 + 299x^8y$
 $+208x^8 + 8x^7y^6 - 20x^7y^5 - 260x^7y^4 - 367x^7y^3 + 716x^7y^2$
 $+495x^7y - 218x^7 + x^6y^7 - 47x^6y^6 - 167x^6y^5 - 123x^6y^4$
 $+1063x^6y^3 + 438x^6y^2 - 1012x^6y - 228x^6 - 19x^5y^7 - 47x^5y^6$
 $+24x^5y^5 + 978x^5y^4 - 16x^5y^3 - 1952x^5y^2 - 100x^5y + 570x^5$
 $-2x^4y^8 - 17x^4y^7 - 10x^4y^6 + 588x^4y^5 - 574x^4y^4$
 $-2288x^4y^3 + 835x^4y^2 + 2037x^4y - 48x^4 - 20x^3y^8 + 4x^3y^7$
 $+391x^3y^6 - 404x^3y^5 - 1566x^3y^4 + 2307x^3y^3 + 3132x^3y^2$
 $-1449x^3y - 1018x^3 - 8x^2y^9 + 80x^2y^8 + 342x^2y^7 - 29x^2y^6$
 $-711x^2y^5 + 2741x^2y^4 + 2615x^2y^3 - 4235x^2y^2 - 2448x^2y + 1392x^2$
 $-xy^{10} + 65xy^9 + 169xy^8 - 37xy^7 - 840xy^6 + 1456xy^5 + 992xy^4 - 6709xy^3$
 $-2065xy^2 + 4702xy + 1107x + 19y^{10} + 19y^9 - 112y^8 - 1044y^7 + 532y^6 + 818y^5$
 $-6213y^4 + 645y^3 + 10019y^2 + 400y - 5488$

B $-2x^4y^{10} - 14x^4y^9 + 16x^4y^8 + 684x^4y^7 - 261x^4y^6$
 $-2372x^4y^5 + 1959x^4y^4 - 1009x^4y^3 - 13335x^4y^2 - 8498x^4y$
 $+692x^4 - 20x^3y^{10} + 12x^3y^9 + 369x^3y^8 - 613x^3y^7 - 2496x^3y^6$
 $+1956x^3y^5 + 863x^3y^4 - 9215x^3y^3 + 3342x^3y^2 + 18310x^3y$
 $+11122x^3 - 8x^2y^{11} + 81x^2y^{10} + 373x^2y^9 + 45x^2y^8 - 455x^2y^7$
 $+4642x^2y^6 + 5972x^2y^5 - 2053x^2y^4 + 13112x^2y^3 + 26973x^2y^2$
 $+5331x^2y - 5568x^2 - xy^{12} + 64xy^{11} + 202xy^{10} + 118xy^9 - 302xy^8$
 $+2994xy^7 + 3318xy^6 - 7665xy^5 + 2259xy^4 + 10284xy^3 - 25464xy^2 - 34124xy$
 $-9828x + 19y^{12} + 34y^{11} - 86y^{10} - 1030y^9 - 374y^8 - 40y^7 - 10501y^6$
 $-7712y^5 + 5611y^4 - 18841y^3 - 21749y^2 + 18327y + 19927$

$i = 6, m = 1$, neighbours 39, 3, 15, 21, 23, 27, 30, 34

W $10x^4 - 10x^3 - 20x - 40$

A $-5x^7 - 10x^6y + 5x^6 - 5x^5y^2 - 10x^5y + 10x^5 - 25x^4y^2 + 30x^4y$
 $-10x^3y^3 + 30x^3y^2 + 25x^3y - 75x^3 + 25x^2y^3 + 40x^2y^2 + 25x^2y$
 $+195x^2 + 20xy^4 + 40xy^3 + 55xy^2 + 40xy - 170x + 5y^5 + 115y^4 - 65y^3 - 375y^2$
 $-230y - 400$

B $-55x^4y^4 + 120x^4y^3 + 370x^4y^2 + 120x^4y + 760x^4 - 10x^3y^5$
 $+95x^3y^4 + 300x^3y^3 + 80x^3y^2 + 505x^3y + 380x^3 + 55x^2y^5$
 $+245x^2y^4 + 135x^2y^3 + 355x^2y^2 + 495x^2y - 480x^2 + 25xy^6 + 185xy^5$
 $+35xy^4 - 15xy^3 + 365xy^2 - 670xy - 1140x + 5y^7 + 125y^6 - 25y^5 - 345y^4$
 $+90y^3 - 860y^2 - 1580y - 920$

$i = 7, m = 1$, neighbours 0, 11, 15, 22, 25, 27

W $x^4 - 6x^3 + 6x^2$

A $x^4y - 2x^4 + x^3y^2 - 5x^3y + 5x^3 + x^2y^2 + 2x^2y + 4x^2 + 2xy^3$
 $-9xy^2 + 5xy - 38x - 5y^3 + y^2 - 19y$

B $-x^4y^3 - 7x^4y^2 - x^4y - 35x^4 + x^3y^4 + 2x^3y^3 + 9x^3y^2$
 $+3x^3y + 18x^3 - x^2y^4 - 7x^2y^3 - 8x^2y^2 - 5x^2y + 4x^2 + 2xy^5$
 $+xy^4 + 11xy^3 + 7xy^2 + 11xy + 12x - 4y^5 - 25y^3 - 16y^2 + 12y - 76$

$i = 8, m = 1$, neighbours 38, 0, 4, 5, 12, 16, 19, 26, 29, 33, 34

W $-7x^3 - 2x^2 - 7x - 12$

A $-2x^9y^2 + 2x^9 + 3x^8y^3 + 8x^8y^2 - 3x^8y - 8x^8 - x^7y^4$
 $-8x^7y^3 - 5x^7y^2 + 8x^7y + 6x^7 + 2x^6y^4 + 13x^6y^3 - 16x^6y^2$
 $-13x^6y + 14x^6 - 5x^5y^4 - 35x^5y^3 + 33x^5y^2 + 35x^5y - 28x^5$
 $-10x^4y^5 + 5x^4y^4 + 48x^4y^3 - 11x^4y^2 - 38x^4y + 6x^4 + 3x^3y^6$
 $+20x^3y^5 - 17x^3y^4 + 5x^3y^3 - 30x^3y^2 - 25x^3y + 44x^3 + x^2y^7$
 $-2x^2y^6 - 13x^2y^5 + 89x^2y^4 - 72x^2y^3 - 43x^2y^2 + 84x^2y$
 $-44x^2 - 2xy^7 + 11xy^6 + 23xy^5 - 164xy^4 + 33xy^3 + 201xy^2 - 54xy - 97x + 11y^7$
 $-7y^6 - 37y^5 + 130y^4 + 190y^3 - 105y^2 - 139y - 41$

B $-3x^4y^7 + 6x^4y^6 - 10x^4y^5 + 29x^4y^4 + 59x^4y^3 + 95x^4y^2$
 $-21x^4y - 125x^4 + 2x^3y^8 + 10x^3y^7 - 11x^3y^6 + 33x^3y^5$

	$-123x^3y^4 - 15x^3y^3 - 47x^3y^2 - 58x^3y + 251x^3 + x^2y^9 - x^2y^8$ $-5x^2y^7 + 56x^2y^6 - 76x^2y^5 + 113x^2y^4 + 287x^2y^3 - 135x^2y^2$ $-249x^2y + 121x^2 - 2xy^9 + 6xy^8 + 4xy^7 - 138xy^6 + 103xy^5 + 51xy^4$ $-729xy^3 + 266xy^2 + 561xy - 200x + 7y^9 - 10y^8 - 7y^7 + 161y^6 - 15y^5 - 222y^4$ $+603y^3 + 116y^2 - 536y - 20$
$i = 9, m = 1$, neighbours 39, 5, 21, 23	
W	5
A	-5
B	$-5x^4 + 5y$
$i = 10, m = 1$, neighbours 16, 18	
W	x^3
A	$-x$
B	$x^2 - xy^2 + 2xy - 5x - y + 2$
$i = 11, m = 1$, neighbours 0, 1, 7, 15, 17, 22, 23	
W	$-8x^4 - 8x^3 + 2x^2 + 24$
A	$-4x^6 - 8x^5y - 5x^4y^2 + 2x^4y + 16x^4 - x^3y^3 + 11x^3y^2 + 18x^3y$ $-52x^3 + 9x^2y^3 + 25x^2y^2 - 14x^2y - 140x^2 + 2xy^4 + 20xy^3 - 24xy^2$ $-28xy - 40x + 11y^4 - 35y^3 - 5y^2 + 26y + 24$
B	$-5x^4y^4 + 24x^4y^3 + 47x^4y^2 + 51x^4y + 164x^4 - x^3y^5 + 22x^3y^4$ $+43x^3y^3 - 2x^3y^2 + 59x^3y + 300x^3 + 11x^2y^5 + 33x^2y^4$ $+37x^2y^3 + 15x^2y^2 + 134x^2y + 232x^2 + 2xy^6 + 18xy^5 + 4xy^4 + 42xy^3$ $+123xy^2 + 108xy + 80x + 10y^6 - 15y^5 - 30y^4 + 10y^3 - 100y^2 - 220y - 336$
$i = 12, m = 1$, neighbours 40, 3, 8, 26, 31	
W	$3x^4 + 12$
A	$-9x^3 - 6x + 12y - 36$
B	$-12x^4y + 15x^4 - 9x^3y^2 - 36x^3 - 18x^2y + 18x^2 - 6xy^2 - 6xy + 12x + 6y^3$ $-18y^2 + 24$
$i = 13, m = 1$, neighbours 0, 3, 15, 16, 20, 22, 25, 33	
W	$-2x^4 - x^3 - 10x^2 - 8x + 3$
A	$x^8 + x^7y - 7x^7 - x^6y^2 - 7x^6y + 19x^6 - x^5y^3 + 3x^5y^2 + 17x^5y$ $-24x^5 + 3x^4y^3 - 9x^4y^2 - 10x^4y + 9x^4 - 6x^3y^3 + 36x^3y^2$ $-33x^3y + 7x^3 + 5x^2y^4 + 21x^2y^3 - 82x^2y^2 + 85x^2y - x^2 + 5xy^5$ $-7xy^4 - 35xy^3 + 111xy^2 - 80xy - 101x + y^6 - 5y^5 - 3y^4 + 50y^3 - 17y^2 - 117y$ -56
B	$x^4y^5 - 8x^4y^4 + 19x^4y^3 + 42x^4y^2 - 57x^4y - 48x^4 - x^3y^5$ $+20x^3y^4 - 49x^3y^3 - 20x^3y^2 + 37x^3y + 94x^3 + 2x^2y^6$ $+10x^2y^5 - 39x^2y^4 + 73x^2y^3 + 29x^2y^2 - 131x^2y + 44x^2 + 3xy^7$ $-23xy^5 + 67xy^4 - 69xy^3 - 73xy^2 + 264xy - 22x + y^8 - 4y^7 + 3y^6 + 43y^5$ $-88y^4 - 41y^3 + 185y^2 - 253y - 215$
$i = 14, m = 1$, neighbours 4, 19, 22, 24, 33	
W	$x^4 + 2x^3 - 4x - 4$
A	$-x^3y - 2x^3 - x^2y - 5x^2 + xy^2 + 2xy - 3x + y^2 + 5y + 2$
B	$-2x^4y^2 + x^4y - 7x^4 - x^3y^3 - 2x^3y^2 + 4x^3y - 10x^3 - x^2y^3$ $-3x^2y^2 + 2x^2y - 4x^2 + xy^4 - 2xy^3 - xy + 4x + 3y^4 + y^3 + 4y + 12$
$i = 15, m = 1$, neighbours 6, 7, 11, 13, 16, 22, 25, 26, 33	
W	$-4x^4 + 12x^3 + 30x^2 - 52x + 24$
A	$-4x^7y + 8x^7 + 4x^6y^2 + 2x^6y - 20x^6 + 3x^5y^3 - 11x^5y^2 + 16x^5y$ $-12x^5 - 2x^4y^4 - x^4y^3 + 15x^4y^2 - 38x^4y + 56x^4 - x^3y^5$ $+4x^3y^4 - 3x^3y^3 - 16x^3y^2 + 42x^3y - 44x^3 - x^2y^5 - 14x^2y^4$ $+46x^2y^3 - 23x^2y^2 + 20x^2y + 36x^2 - xy^6 - 6xy^5 + 27xy^4 - 77xy^3$ $+111xy^2 - 66xy + 224x + 2y^6 - 11y^5 - 18y^4 + 53y^3 + 20y^2 + 300y - 640$
B	$-4x^4y^6 + 5x^4y^5 - 5x^4y^4 + 10x^4y^3 + 60x^4y^2 + 244x^4y$ $+456x^4 - x^3y^7 - 6x^3y^6 + 2x^3y^5 + 14x^3y^4 - 67x^3y^3$ $-82x^3y^2 - 16x^3y + 612x^3 - 5x^2y^7 - 6x^2y^6 + 38x^2y^5$ $+18x^2y^4 - 31x^2y^3 - 180x^2y^2 - 472x^2y - 1496x^2 - xy^8 - 13xy^7$ $+8xy^6 + 13xy^5 + 128xy^4 + 199xy^3 + 86xy^2 + 408xy + 1296x - 11y^7 - 26y^6$ $+13y^5 - 22y^4 + 168y^3 - 92y^2 - 104y + 992$
$i = 16, m = 1$, neighbours 3, 4, 5, 8, 10, 13, 15, 19, 20, 30, 34	
W	$-226x^4 + 94x^3 + 718x^2 - 1444x + 408$
A	$-2x^7y^3 - 4x^7y^2 + 2x^7y + 4x^7 - 7x^6y^4 - 13x^6y^3 + 9x^6y^2$

$$\begin{aligned}
& +13x^6y - 2x^6 - 9x^5y^5 - 22x^5y^4 + 12x^5y^3 + 44x^5y^2 - 3x^5y \\
& - 22x^5 - 5x^4y^6 - 30x^4y^5 - x^4y^4 + 97x^4y^3 + 4x^4y^2 - 67x^4y \\
& + 2x^4 - x^3y^7 - 29x^3y^6 - 12x^3y^5 + 135x^3y^4 + 42x^3y^3 \\
& - 144x^3y^2 - 29x^3y - 51038x^3 - 15x^2y^7 - 8x^2y^6 + 128x^2y^5 \\
& + 98x^2y^4 - 205x^2y^3 - 82x^2y^2 + 92x^2y + 42480x^2 - 3xy^8 + 93xy^6 \\
& + 116xy^5 - 177xy^4 - 124xy^3 + 99xy^2 + 8xy + 315688x + 10y^8 + 186y^7 - 79y^6 \\
& + 9y^5 + 1419y^4 - 11811y^3 + 54862y^2 - 147008y - 562448 \\
B \quad & - 5x^4y^8 - 115x^4y^7 + 192x^4y^6 - 103x^4y^5 - 1540x^4y^4 \\
& + 11848x^4y^3 - 54939x^4y^2 + 146994x^4y - 225144x^4 - x^3y^9 \\
& - 63x^3y^8 + 68x^3y^7 + 209x^3y^6 - 1813x^3y^5 + 8783x^3y^4 \\
& - 31556x^3y^3 + 37119x^3y^2 + 18038x^3y - 509932x^3 - 21x^2y^9 \\
& + 14x^2y^8 + 248x^2y^7 - 1399x^2y^6 + 5272x^2y^5 - 14547x^2y^4 \\
& + 24921x^2y^3 + 92916x^2y^2 - 431268x^2y + 1181168x^2 - 3xy^{10} + 5xy^9 \\
& + 201xy^8 - 893xy^7 + 2675xy^6 - 4613xy^5 - 4669xy^4 + 101217xy^3 - 245116xy^2 \\
& + 633952xy - 478608x + 8y^{10} + 162y^9 - 403y^8 + 891y^7 + 423y^6 - 12985y^5 \\
& + 80828y^4 - 303004y^3 - 3684y^2 + 226816y - 1041664 \\
i = 17, m = 1, \text{ neighbours } 3, 5, 11, 22, 23, 30, 32, 33 \\
W \quad & 96x^4 + 380x^3 + 692x^2 + 562x + 100 \\
A \quad & -x^6y^2 + 3x^6y - 2x^6 - x^5y^3 + 2x^5y^2 + x^5y - 2x^5 + 3x^4y^3 \\
& - 4x^4y^2 - 9x^4y + 10x^4 + 4x^3y^4 - x^3y^3 - 21x^3y^2 + 10x^3y \\
& - 9208x^3 + 2x^2y^4 - 15x^2y^3 + 22x^2y^2 + 9x^2y - 72978x^2 - 3xy^5 \\
& - 14xy^4 + 33xy^3 + 43xy^2 - 57xy - 277266x + y^6 - 20y^5 + 125y^4 - 1169y^3 \\
& + 11814y^2 - 78833y - 214478 \\
B \quad & -3x^4y^5 + 53x^4y^4 - 604x^4y^3 + 5937x^4y^2 - 39413x^4y + 209662x^4 \\
& + 4x^3y^6 - 34x^3y^5 + 242x^3y^4 - 2317x^3y^3 + 13765x^3y^2 \\
& - 60856x^3y + 252948x^3 + 7x^2y^6 - 100x^2y^5 + 965x^2y^4 - 9174x^2y^3 \\
& + 16589x^2y^2 - 29113x^2y + 155542x^2 - 5xy^7 + 38xy^6 - 352xy^5 + 3613xy^4 \\
& - 35679xy^3 - 1960xy^2 + 90001xy - 74376x + y^8 - 20y^7 + 167y^6 - 1509y^5 \\
& + 14742y^4 - 107809y^3 - 3528y^2 + 54312y - 423956 \\
i = 18, m = 1, \text{ neighbours } 2, 10, 23, 31, 34 \\
W \quad & 4x^3 - 6x^2 + 12x \\
A \quad & xy^2 + 2xy - 16x + y^2 - 4 \\
B \quad & -x^4y - 26x^4 + 11x^3y + 40x^3 + x^2y^3 - 4x^2y^2 - 10x^2y - 32x^2 + xy^4 \\
& + 5xy^3 + 16xy + 40x - y^4 - 4y^3 - 4y^2 + 4y + 8 \\
i = 19, m = 1, \text{ neighbours } 8, 14, 16, 26, 27, 30, 31 \\
W \quad & -4x^5 + 4x^4 - 4x^3 + 4x^2 + 20x \\
A \quad & x^6 - x^5y^2 - 3x^5y - 11x^5 + x^4y^3 + 4x^4y^2 - 4x^4y + 24x^4 \\
& - x^3y^3 - 5x^3y^2 - 8x^3y - 52x^3 - 3x^2y^2 - 3x^2y + 69x^2 - 2xy^3 \\
& + 3xy^2 - 36xy - 16x - y^5 - 4y^4 + 2y^3 + 5y^2 + 133y - 16 \\
B \quad & x^8y^2 + 4x^8y - 4x^8 - x^7y^3 - 5x^7y^2 + 4x^7 - x^6y^3 - 4x^6y^2 \\
& + 4x^6y + x^5y^4 + 5x^5y^3 + x^5y^2 - 13x^5 + x^4y^4 + 2x^4y^3 \\
& - 12x^4y^2 + 17x^4y + 116x^4 + x^3y^5 + 5x^3y^4 + 20x^3y^3 + 11x^3y^2 \\
& - 52x^3y - 36x^3 - x^2y^6 - 3x^2y^5 + 11x^2y^4 + 3x^2y^3 + 7x^2y^2 \\
& - 88x^2y + 80x^2 - 11xy^4 + 42xy^3 + 61xy^2 + 137xy + 64x + y^7 + 3y^6 - 7y^5 - y^4 \\
& - 81y^3 + 56y^2 - 68y - 660 \\
i = 20, m = 1, \text{ neighbours } 38, 0, 2, 5, 13, 16 \\
W \quad & -2x^3 + 2x^2 + 4x - 4 \\
A \quad & 4x + 2y^2 + 4y - 4 \\
B \quad & -8x^4 - 2x^3y^2 - 4x^3y + 4x^3 + 2x^2y + 16x^2 + xy^3 + 3xy - 16x - y^4 - 2y^3 \\
& + 3y^2 - 10y - 16 \\
i = 21, m = 1, \text{ neighbours } 41, 3, 6, 9 \\
W \quad & 5 \\
A \quad & -5 \\
B \quad & 5x^3 - 5xy + 5y - 5 \\
i = 22, m = 16, \text{ neighbours } 38, 4, 7, 11, 13, 14, 15, 17, 24, 26, 30, 33 \\
W \quad & -416x^5 - 3328x^4 + 32x^3 + 192x^2 - 1920x + 4864 \\
A \quad & -512x^8y^5 + 13952x^8y^4 - 23872x^8y^3 + 445536x^8y^2 - 2480336x^8y \\
& + 21632280x^8 + 7168x^7y^3 - 4096x^7y^2 - 17408x^7y - 6144x^7 \\
& + 9728x^6y^4 - 6656x^6y^3 - 39936x^6y^2 + 11264x^6y + 34816x^6 \\
& + 6400x^5y^5 - 17920x^5y^4 - 58112x^5y^3 + 52224x^5y^2 + 141312x^5y \\
& - 152207412x^5 + 1536x^4y^7 - 39808x^4y^6 + 31680x^4y^5 - 1032736x^4y^4
\end{aligned}$$

$$\begin{aligned}
& +6935536x^4y^3 - 53145992x^4y^2 - 64456992x^4y - 313710510x^4 \\
& +256x^3y^7 - 18688x^3y^6 - 49152x^3y^5 + 167168x^3y^4 + 267008x^3y^3 \\
& -244736x^3y^2 - 470016x^3y - 11204608x^3 - 6144x^2y^7 - 21760x^2y^6 \\
& +158208x^2y^5 + 203520x^2y^4 - 486400x^2y^3 - 604160x^2y^2 + 26624x^2y \\
& +487424x^2 - 768xy^8 - 8192xy^7 + 115200xy^6 + 90112xy^5 - 406784xy^4 \\
& -824064xy^3 + 462266652xy^2 - 29164992xy - 128633442x - 1024y^9 + 17920y^8 \\
& +39168y^7 + 483968y^6 - 2541632y^5 + 21636704y^4 + 155522712y^3 - 162865186y^2 \\
& -133302882y - 5914624 \\
B \quad & 512x^{11}y^5 - 13952x^{11}y^4 + 23872x^{11}y^3 - 443488x^{11}y^2 \\
& +2478288x^{11}y - 21636376x^{11} - 512x^{10}y^5 + 13952x^{10}y^4 - 23872x^{10}y^3 \\
& +443488x^{10}y^2 - 2478288x^{10}y + 21636376x^{10} - 512x^9y^6 + 14464x^9y^5 \\
& -37824x^9y^4 + 467360x^9y^3 - 2921776x^9y^2 + 24114664x^9y - 21636376x^9 \\
& +1024x^8y^6 - 28416x^8y^5 + 61696x^8y^4 - 910848x^8y^3 \\
& +5400064x^8y^2 - 45751040x^8y + 173726028x^8 - 1024x^7y^7 + 25856x^7y^6 \\
& +21888x^7y^5 + 414784x^7y^4 - 2538080x^7y^3 + 21385424x^7y^2 \\
& +153459280x^7y + 137119330x^7 + 512x^6y^8 - 13952x^6y^7 + 26432x^6y^6 \\
& -525024x^6y^5 + 2918544x^6y^4 - 24402872x^6y^3 + 22591664x^6y^2 \\
& -317272156x^6y - 223664834x^6 + 512x^5y^8 - 11904x^5y^7 - 48320x^5y^6 \\
& +110240x^5y^5 - 380464x^5y^4 + 3017448x^5y^3 - 176050944x^5y^2 \\
& +180152826x^5y + 223664834x^5 - 512x^4y^9 + 13440x^4y^8 - 14528x^4y^7 \\
& +573344x^4y^6 - 3022640x^4y^5 + 24615912x^4y^4 - 25322648x^4y^3 \\
& +31732288x^4y^2 - 78838188x^4y - 95269984x^4 + 1024x^3y^9 - 22528x^3y^8 \\
& +57600x^3y^7 - 1082624x^3y^6 + 4481408x^3y^5 - 42668352x^3y^4 \\
& +145680300x^3y^3 - 112864568x^3y^2 - 325475938x^3y + 701277488x^3 \\
& -1024x^2y^{10} + 25600x^2y^9 + 28416x^2y^8 + 352896x^2y^7 \\
& -2923200x^2y^6 + 24085280x^2y^5 + 137375352x^2y^4 + 108765698x^2y^3 \\
& +403579140x^2y^2 + 666955586x^2y + 585927048x^2 - 256xy^{10} + 21760xy^9 \\
& -61696xy^8 + 465536xy^7 - 1509568xy^6 + 19507488xy^5 - 283654692xy^4 \\
& +13385526xy^3 - 77560482xy^2 - 1366176882xy - 540715400x + 1024y^{11} - 18432y^{10} \\
& -40192y^9 - 474240y^8 + 2482240y^7 - 21811808y^6 - 157507736y^5 + 173039906y^4 \\
& +46821602y^3 - 616143456y^2 + 651460744y + 533211528 \\
i = 23, m = 1, \text{ neighbours } 3, 6, 9, 11, 17, 18 \\
W \quad & -2x^5 - 2x^4 + 5x^3 - 3x^2 - 11x + 3 \\
A \quad & 4x^5 - 8x^4y + 16x^3 + 8x^2 - 44xy - 24x + 9y^3 + 12y^2 - 12y - 24 \\
B \quad & -8x^8 + 8x^7y - 8x^7 - 8x^6y - 8x^6 + 8x^5y^2 - 8x^5 - 8x^4y^2 + 32x^4y \\
& -65x^4 - 5x^3y^3 - 12x^3y^2 + 6x^3 + 8x^2y^4 - 8x^2y^3 + 16x^2y^2 \\
& +35x^2y + 40x^2 - 5xy^4 - 34xy^3 - 6xy^2 + 10xy - 162x + 8y^5 + 7y^4 - 17y^3 \\
& +17y^2 + 39y - 87 \\
i = 24, m = 2, \text{ neighbours } 3, 4, 5, 14, 22, 31 \\
W \quad & -4x^5 - 12x^4 - 16x^3 - 8x^2 \\
A \quad & -14x^5 - 171x^4 - 272x^3 - 448x^2 + 6xy^2 - 36xy - 333x + 135y^2 + 333y \\
B \quad & -2x^8 + 77x^7 - 2x^6y - 77x^6 + 79x^5y + 77x^5 - 4x^4y^2 - 134x^4y - 196x^4 \\
& +22x^3y^3 - 40x^3y^2 - 301x^3y - 16x^3 + 4x^2y^4 + 161x^2y^3 \\
& -82x^2y^2 - 183x^2y + 620x^2 + 22xy^4 + 91xy^3 - 199xy^2 - 301xy + 444x + 4y^5 \\
& +135y^4 + 313y^3 + 556y + 1332 \\
i = 25, m = 1, \text{ neighbours } 7, 13, 15, 27, 32 \\
W \quad & 3x^3 - 8x^2 + x + 3 \\
A \quad & -5x^4 - 4x^2 - 8xy + 7x + 5y^2 - 3y - 11 \\
B \quad & 5x^7 + 5x^6 - 5x^5y - 10x^4y - 19x^4 - 5x^3y^2 - 10x^3y + 46x^3 + 4x^2y^3 \\
& +x^2y^2 + 14x^2y + 17x^2 + 8xy^3 + 8xy^2 - 17xy - 7x - 4y^4 + 9y^3 - y^2 - 27y + 35 \\
i = 26, m = 1, \text{ neighbours } 5, 8, 12, 15, 19, 22, 30, 33 \\
W \quad & 2x^5 + 12x^4 + 20x^3 - 40x^2 - 32x + 48 \\
A \quad & -2x^6y + 2x^6 + 2x^5y^3 - x^5y^2 + 25x^5y + 84x^5 - 2x^4y^4 - 12x^4y^3 \\
& -27x^4y^2 - 183x^4y - 508x^4 + x^3y^4 + x^3y^3 + 9x^3y^2 - 17x^3y \\
& -218x^3 + 16x^2y^3 - 27x^2y^2 + 19x^2y - 478x^2 - xy^5 + 8xy^4 - 18xy^3 \\
& -35xy^2 + 204xy + 36x + 2y^6 + 4y^5 + y^4 + 110y^3 - 203y^2 - 1720y + 2908 \\
B \quad & -2x^8y^3 - 2x^8y^2 - 20x^8y - 90x^8 + 2x^7y^4 + 12x^7y^3 + 30x^7y^2 \\
& +190x^7y + 450x^7 + 2x^6y^4 - 2x^6y^3 + 16x^6y^2 + 50x^6y - 180x^6 \\
& -2x^5y^5 - 8x^5y^4 - 8x^5y^3 - 132x^5y^2 - 90x^5y + 1074x^5 \\
& -2x^4y^5 + 10x^4y^4 - 16x^4y^3 + 22x^4y^2 + 196x^4y + 280x^4 \\
& -2x^3y^6 - 4x^3y^5 - 45x^3y^4 - 149x^3y^3 - 102x^3y^2 - 144x^3y \\
& +1072x^3 + 2x^2y^7 + 11x^2y^6 + 17x^2y^5 + 211x^2y^4 + 528x^2y^3
\end{aligned}$$

$$\begin{aligned}
& -37x^2y^2 + 602x^2y + 1120x^2 + xy^7 - xy^6 + 22xy^5 + 79xy^4 + 36xy^3 \\
& -99xy^2 - 1778xy - 1464x - 2y^8 - 6y^7 - 13y^6 - 154y^5 - 121y^4 + 1482y^3 \\
& -384y^2 + 44y + 6968
\end{aligned}$$

$i = 27, m = 1$, neighbours 6, 7, 19, 25, 31

$$\begin{aligned}
W & -2x^4 - 8x^2 - 8 \\
A & -x^5 + x^4y - 3x^4 - 6x^3 + x^2y + x^2 + xy^2 - 2xy - 8x - y^3 + 7y + 14 \\
B & x^8 - x^7y + 3x^7 - x^6y - 2x^6 + x^5y^2 - x^5y - 5x^5 + x^4y^2 + 2x^4y \\
& -26x^4 + x^3y^3 + 4x^3 - x^2y^4 + 3x^2y^3 + x^2y^2 + 2x^2y - 36x^2 \\
& -xy^4 + 3xy^3 + xy^2 + 3xy + 16x + y^5 - 3y^4 - 8y^2 + 16y - 60
\end{aligned}$$

$i = 28, m = 2$, neighbours 42, 4, 31

$$\begin{aligned}
W & -2x^4 - 4x^3 - 6x^2 - 6x \\
A & -2x^4 - 4x^3 - 16x^2 - 10x + 2y^2 - 14y - 16 \\
B & 2x^7 - 2x^6 - 2x^5y + 4x^4y - 28x^4 + 10x^3y - 28x^3 + 2x^2y^3 - 4x^2y^2 \\
& + 10x^2y - 20x^2 + 2xy^3 - 6xy^2 - 8xy - 8x - 2y^4 + 16y^3 + 48
\end{aligned}$$

$i = 29, m = 2$, neighbours 41, 5, 8, 33

$$\begin{aligned}
W & 2x^5 - 2x^4 + 8x^3 - 10x^2 + 2x \\
A & -6x^5 + 15x^4 - 36x^3 + 72x^2 + 6xy^2 + 4xy - 55x - 7y^2 - 47y + 54 \\
B & 2x^8 - 5x^7 + 2x^6y - 7x^6 - 3x^5y - 2x^5 - 4x^4y^2 - 22x^4y - 52x^4 \\
& + 6x^3y^3 + 41x^3y + 88x^3 - 13x^2y^3 - 18x^2y^2 - 25x^2y - 52x^2 + 6xy^4 \\
& + 17xy^3 + 23xy^2 + 114xy + 24x - 7y^4 - 54y^3 + 40y^2 + 12y - 216
\end{aligned}$$

$i = 30, m = 1$, neighbours 1, 6, 16, 17, 19, 22, 26, 32

$$\begin{aligned}
W & -8x^5 - 14x^4 + 14x^3 + 32x^2 + 4x - 8 \\
A & -2x^8 + 3x^7y - 4x^7 + 9x^6y + 8x^6 - x^5y^3 + 25x^5y^2 - 113x^5y + 183x^5 \\
& -29x^4y^3 + 196x^4y^2 - 584x^4y + 453x^4 - x^3y^4 - 9x^3y^3 \\
& + 18x^3y^2 - 13x^3y + 8x^3 - 7x^2y^4 - 13x^2y^3 + 27x^2y^2 - 85x^2y \\
& + 1046x^2 + 2xy^5 - 13xy^4 - 25xy^3 + 86xy^2 - 218xy - 716x + 36y^5 - 68y^4 \\
& - 183y^3 + 970y^2 - 424y - 3936 \\
B & -26x^8y^2 + 112x^8y - 223x^8 + 26x^7y^3 - 190x^7y^2 + 559x^7y - 669x^7 \\
& -26x^6y^3 + 60x^6y^2 + x^6y - 446x^6 + 26x^5y^4 - 138x^5y^3 \\
& + 153x^5y^2 + 561x^5y - 1502x^5 - 26x^4y^4 + 60x^4y^3 - 103x^4y^2 \\
& - 57x^4y + 544x^4 + x^3y^6 - 33x^3y^5 + 56x^3y^4 + 105x^3y^3 \\
& - 765x^3y^2 + 927x^3y - 700x^3 + 31x^2y^6 - 217x^2y^5 + 571x^2y^4 \\
& - 307x^2y^3 - 1008x^2y^2 + 2758x^2y - 2076x^2 + 2xy^7 - 18xy^6 - 29xy^5 \\
& + 223xy^4 - 517xy^3 - 748xy^2 + 2264xy - 1464x + 37y^7 - 140y^6 + 92y^5 + 920y^4 \\
& - 2532y^3 + 404y^2 + 3072y - 7840
\end{aligned}$$

$i = 31, m = 1$, neighbours 0, 12, 18, 19, 24, 27, 28

$$\begin{aligned}
W & -8x^5 + 2x^4 - 18x^3 - 12x^2 - 20x - 24 \\
A & -4x^7 + x^5y^2 - 8x^5y - 73x^5 + 3x^4y^2 + 17x^4y + 61x^4 + x^3y^3 \\
& + 8x^3y^2 + 4x^3y - 284x^3 + 4x^2y^3 + x^2y^2 - 202x^2 - 2xy^4 + 10xy^3 \\
& + 29xy^2 + 52xy - 324x - 7y^4 - 6y^3 + 37y^2 + 266y - 568 \\
B & 2x^8y + 5x^8 - 2x^7y^2 - 15x^7y - 25x^7 + 2x^6y^2 + x^6y - 10x^6 \\
& - 2x^5y^3 - 11x^5y^2 + 7x^5y + 152x^5 + 2x^4y^3 - 7x^4y^2 - 119x^4y \\
& - 252x^4 - x^3y^5 + 11x^3y^4 + 67x^3y^3 - 13x^3y^2 - 55x^3y - 188x^3 \\
& - 5x^2y^5 - 23x^2y^4 - 50x^2y^3 - 91x^2y^2 - 362x^2y - 508x^2 - 2xy^6 \\
& + 5xy^5 + 42xy^4 + 126xy^3 - 77xy^2 - 304xy + 168x - 8y^6 - 23y^5 - 5y^4 + 232y^3 \\
& - 128y^2 - 108y + 848
\end{aligned}$$

$i = 32, m = 1$, neighbours 5, 17, 25, 30

$$\begin{aligned}
W & x^3 - 3x - 3 \\
A & -2x - 3 \\
B & x^4 + 2x^3y + x^2y^2 - 2x^2y + 7x^2 + xy + 3x + y^3 + 3y - 7
\end{aligned}$$

$i = 33, m = 3456$, neighbours 3, 4, 5, 8, 13, 14, 15, 17, 22, 26, 29

$$\begin{aligned}
W & -393984x^5 - 506304x^4 + 737856x^3 - 705024x^2 + 898560x - 3525120 \\
A & -995328x^8y^5 - 46199808x^8y^4 - 154116864x^8y^3 - 173328192x^8y^2 \\
& - 899845680x^8y - 5217438836x^8 + 47775744x^7y^2 - 47775744x^7 \\
& + 238878720x^6y^3 + 119439360x^6y^2 - 238878720x^6y - 119439360x^6 \\
& + 489701376x^5y^4 + 561364992x^5y^3 - 680804352x^5y^2 - 561364992x^5y \\
& - 354798101852x^5 + 4976640x^4y^7 + 243938304x^4y^6 + 1905672960x^4y^5 \\
& + 4396744512x^4y^4 + 6933209136x^4y^3 + 37589383156x^4y^2 \\
& + 76606120340x^4y - 620323490014x^4 + 310542336x^3y^6 + 1218281472x^3y^5 \\
& - 1457160192x^3y^4 - 3272638464x^3y^3 + 1074954240x^3y^2 + 2054356992x^3y \\
& + 325134839808x^3 + 95551488x^2y^7 + 824131584x^2y^6 - 1552711680x^2y^5
\end{aligned}$$

	$-3810115584x^2y^4 + 2018525184x^2y^3 + 3415965696x^2y^2 - 561364992x^2y$ $+19119255520x^2 + 11943936xy^8 + 358318080xy^7 - 1672151040xy^6$ $-3535405056xy^5 + 4299816960xy^4 + 4216209408xy^3 + 995783359180xy^2$ $+2440693048748xy + 1128823057070x - 3981312y^9 - 282341376y^8 - 3013051392y^7$ $-5738893056y^6 - 4877327424y^5 - 24640428272y^4 + 40573356488y^3$ $+1711649923232y^2 - 909885713060y - 3106617753600$
B	$995328x^{11}y^5 + 46199808x^{11}y^4 + 154116864x^{11}y^3 + 173328192x^{11}y^2$ $+899845680x^{11}y + 5217438836x^{11} + 995328x^{10}y^5 + 46199808x^{10}y^4$ $+154116864x^{10}y^3 + 173328192x^{10}y^2 + 899845680x^{10}y + 5217438836x^{10}$ $+1990656x^9y^6 + 93394944x^9y^5 + 354433536x^9y^4 + 500773248x^9y^3$ $+1973019552x^9y^2 + 11334723352x^9y + 5217438836x^9 + 3981312x^8y^6$ $+185794560x^8y^5 + 662667264x^8y^4 + 847429632x^8y^3$ $+3772710912x^8y^2 + 21769601024x^8y + 204983251408x^8 - 995328x^7y^7$ $-54162432x^7y^6 - 531677952x^7y^5 - 1775861568x^7y^4$ $-3519406128x^7y^3 - 13802829812x^7y^2 - 48938276128x^7y + 427000363694x^7$ $+995328x^6y^8 + 53167104x^6y^7 + 471543552x^6y^6 + 970966080x^6y^5$ $+1003642608x^6y^4 + 9859921988x^6y^3 + 30429685004x^6y^2$ $+364627609408x^6y + 447870119038x^6 - 995328x^5y^8 - 55157760x^5y^7$ $-591812352x^5y^6 - 2580757056x^5y^5 - 6035169648x^5y^4$ $-17745737636x^5y^3 - 67446867252x^5y^2 + 1218628336796x^5y$ $+447870119038x^5 + 1990656x^4y^9 + 105338880x^4y^8 + 887929344x^4y^7$ $+1350119808x^4y^6 - 573471840x^4y^5 + 13684674328x^4y^4$ $+43113632372x^4y^3 - 337020711296x^4y^2 - 682352801136x^4y$ $-1230854737456x^4 + 3981312x^3y^9 + 377892864x^3y^8 + 1432470528x^3y^7$ $-144158976x^3y^6 + 1790041152x^3y^5 + 16336001264x^3y^4$ $+253034879892x^3y^3 + 463269185708x^3y^2 + 506508952914x^3y$ $+2467575088960x^3 - 995328x^2y^{10} - 17335296x^2y^9 - 229678848x^2y^8$ $-2529400896x^2y^7 - 4649425392x^2y^6 - 8109967748x^2y^5$ $-802775228x^2y^4 + 163557785778x^2y^3 - 1045881700868x^2y^2$ $-1934244703582x^2y - 444582466888x^2 + 6967296xy^{10} + 308468736xy^9$ $+79259904xy^8 + 717686208xy^7 + 7040589456xy^6 + 11710359644xy^5$ $+499318904428xy^4 + 784286605834xy^3 - 31418578784xy^2 + 3183020638498xy$ $+4723188378296x - 1990656y^{11} - 135198720y^{10} - 1500553728y^9 - 2937128832y^8$ $-3063066144y^7 - 18149241976y^6 + 9005967076y^5 + 845831428048y^4$ $-504462591794y^3 - 1472066612336y^2 + 3423395397952y - 1819771426120$
$i = 34, m = 8, \text{neighbours } 41, 3, 5, 6, 8, 16, 18$	
W	$10x^5 - 2x^4 - 2x^3 + 26x^2 - 24x + 32$
A	$140x^5 - 417x^4 + 36x^3 - 528x^2 - 1200xy^2 + 2900xy + 351x + 320y^3 + 4708y^2 - 66y$ -256
B	$-240x^8 + 697x^7 + 480x^6y - 697x^6 - 1874x^5y + 697x^5 + 240x^4y^2$ $+508x^4y - 468x^4 - 460x^3y^3 - 1200x^3y^2 - 1613x^3y - 1108x^3$ $+397x^2y^3 + 504x^2y^2 + 985x^2y + 3412x^2 + 920xy^4 + 687xy^3 + 1762xy^2$ $+2303xy - 1540x - 2354y^4 - 767y^3 - 512y^2 - 8776y + 132$

Table 2: The data of the identities (3).

$\{i, j\} = \{0, 2\}, k_{ij} = 2^2 \cdot 5$	
α	$-x^4 + 5$
β	$x^7 + x^6 + x^5y + x^5 + 2x^4y + x^4 - 4x^3 - 4xy - 4y$
γ	$-x^5y - 2x^5 - x^4y^2 - 2x^4y - x^4 + 4xy - x + 5y^2 + 4y$
$\{i, j\} = \{0, 5\}, k_{ij} = 2^2 \cdot 3 \cdot 5$	
α	$-x^4 + 11$
β	$x^7 + x^6 + x^5y + x^5 + 2x^4y + x^4 - 7x^3 + x^2 - 7xy + x - 6y + 1$
γ	$-x^5y - 3x^5 - x^4y^2 - 3x^4y - 4x^4 + 7xy + x + 11y^2 + 17y + 16$
$\{i, j\} = \{0, 7\}, k_{ij} = 2^2 \cdot 13$	
α	$81x + 9y^3 + 13$
β	$-81x^4 + 9x^3y^3 + 166x^3 + 9x^2y^4 - 45x^2y^3 + 49x^2y - 362x^2 - 9xy^4$ $+63xy^3 - 49xy + 298x + 9y^5 - 81y^3 + 49y^2 - 324$
γ	$9x^4y^3 + 130x^4 - 27x^3y^3 - 264x^3 + 54x^2y^3 + 330x^2 - 72xy^3 - 311x$ $+81y^3 + 324$

$$\begin{aligned}
&\{i, j\} = \{0, 8\}, k_{ij} = 5 \\
&\alpha \quad 1 \\
&\beta \quad -x^3 - x^2 - xy - x - 2y \\
&\gamma \quad x + y^2 + y - 1 \\
&\{i, j\} = \{0, 11\}, k_{ij} = 2^3 \cdot 5^2 \cdot 13 \\
&\alpha \quad -9x^4 + 209 \\
&\beta \quad 9x^7 + 9x^6 + 9x^5y + 9x^5 + 18x^4y + 9x^4 - 128x^3 + 70x^2 - 128xy + 70x - 58y + 70 \\
&\gamma \quad -9x^5y - 36x^5 - 9x^4y^2 - 36x^4y - 81x^4 + 128xy + 35x + 209y^2 + 476y + 882 \\
&\{i, j\} = \{0, 13\}, k_{ij} = 2^4 \\
&\alpha \quad -40x^5 - 6y^2 + 22y - 36 \\
&\beta \quad 48x^8 + 12x^7y + 60x^7 + 60x^6y + 60x^6 - 2x^5y^3 + 2x^5y^2 + 170x^5y \\
&\quad + 102x^5 + 12x^3y^2 - 102x^3y - 96x^3 + 12x^2y^2 - 177x^2y - 171x^2 \\
&\quad + 12xy^3 - 90xy^2 - 273xy - 171x + 14y^3 - 189y^2 - 258y + 153 \\
&\gamma \quad 4x^9 + 4x^8y + 4x^8 - 2x^7y^2 - 4x^7y - 2x^7 - 2x^6y^2 + 28x^6y + 22x^6 \\
&\quad + 2x^5y^3 - 44x^5y^2 - 120x^5y - 62x^5 - 115x^4 + 20x^3y + 20x^3 \\
&\quad - 10x^2y^2 - 20x^2y - 10x^2 + 54xy^2 + 108xy + 167x - 6y^4 + 14y^3 + 139y^2 + 95y \\
&\quad - 160 \\
&\{i, j\} = \{0, 20\}, k_{ij} = 2^3 \cdot 5 \\
&\alpha \quad -2x^4 + 10 \\
&\beta \quad 2x^7 + 2x^6 + x^5y - x^4y^2 + 2x^4y + x^4 - 8x^3 - 4xy - x + 5y^2 - 4y \\
&\gamma \quad x^5y + 2x^5 + x^4y^2 + 2x^4y + x^4 - 4xy + x - 5y^2 - 4y \\
&\{i, j\} = \{0, 31\}, k_{ij} = 2^5 \cdot 5^3 \\
&\alpha \quad 20 \\
&\beta \quad -2x^9 - 2x^8 - 6x^7 - 2x^6y - 6x^6 + 2x^5y^2 + 6x^5y + 51x^4 - 4x^3y^2 \\
&\quad + 22x^3y + 43x^3 - 4x^2y^2 + 22x^2y + 145x^2 - 4xy^3 + 18xy^2 + 65xy + 750x \\
&\quad + 4y^4 - 18y^3 - 65y^2 - 165y - 1210 \\
&\gamma \quad 2x^9 - 2x^5y^2 - 14x^5y - 6x^5 - 51x^4 - 605x - 4y^4 + 10y^3 + 105y^2 + 375y \\
&\quad + 1960 \\
&\{i, j\} = \{1, 4\}, k_{ij} = 3 \\
&\alpha \quad -1 \\
&\beta \quad x^3 - x^2 + xy - 2y + 1 \\
&\gamma \quad -xy + x - y^2 + 3y - 6 \\
&\{i, j\} = \{1, 11\}, k_{ij} = 3^2 \cdot 5 \\
&\alpha \quad x^4 + 4 \\
&\beta \quad -x^7 + x^6 - x^5y + 2x^4y - x^4 - 7x^3 - 2x^2 - 7xy + 9x + 5y - 7 \\
&\gamma \quad x^5y + 2x^5 + x^4y^2 + 3x^4 + 7xy - x + 4y^2 + 3y + 18 \\
&\{i, j\} = \{1, 30\}, k_{ij} = 3^2 \cdot 5^4 \\
&\alpha \quad 10y - 25 \\
&\beta \quad -x^9 + x^8 + 2x^7 + x^6y - 3x^6 + x^5y^2 - 4x^5y + 3x^5 + 62x^4 - 2x^3y^2 \\
&\quad + 5x^3y - 44x^3 + 2x^2y^2 - 5x^2y - 142x^2 - 2xy^3 + 5xy^2 - 44xy - 699x - 2y^4 \\
&\quad + 7y^3 - 39y^2 + 205y + 2469 \\
&\gamma \quad x^9 - x^5y^2 + 5x^5y - 2x^5 - 62x^4 + 885x + 2y^4 - 3y^3 + 27y^2 - 298y - 1628 \\
&\{i, j\} = \{2, 3\}, k_{ij} = 2^2 \\
&\alpha \quad 1 \\
&\beta \quad -x^4 - x + y^2 \\
&\gamma \quad x^4 \\
&\{i, j\} = \{2, 18\}, k_{ij} = 2 \cdot 5 \\
&\alpha \quad 2 \\
&\beta \quad -x^4 - x^3 - x^2 - 3x + 2y^2 - 1 \\
&\gamma \quad x^4 + x^3 + x^2 + x + 1 \\
&\{i, j\} = \{2, 20\}, k_{ij} = 2^2 \cdot 5 \\
&\alpha \quad -x^4 + 5 \\
&\beta \quad x^7 + x^6 + 2x^5 - x^4y^2 + x^4 - 4x^3 - 5x + 5y^2 \\
&\gamma \quad x^7 + x^6 + x^5 + x^4 - 4x^3 \\
&\{i, j\} = \{3, 4\}, k_{ij} = 5 \\
&\alpha \quad 1 \\
&\beta \quad x^4 - x^3y + x^2y^2 - xy^3 + y^4 - y \\
&\gamma \quad -y^4 + y^3 + y - 1 \\
&\{i, j\} = \{3, 6\}, k_{ij} = 2^3
\end{aligned}$$

α	2
β	$2x^4 - 2x^3y + 2x^2y^2 - 2xy^3 + y^4 + y^2 - y$
γ	$-y^4 + y^2 + y$
$\{i, j\}$	$\{3, 12\}, k_{ij} = 2^2 \cdot 3$
α	3
β	$3x^4 - 3x^3y + 3x^2y^2 - 3xy^3 + y^4 + 2y^2 - y$
γ	$-y^4 + y^2 + y$
$\{i, j\}$	$\{3, 13\}, k_{ij} = 5$
α	1
β	$x^4 - x^3y + x^2y^2 - xy^3 - y^4 - 2y^3 + y + 2$
γ	$y^4 + y^3 - y - 1$
$\{i, j\}$	$\{3, 16\}, k_{ij} = 2 \cdot 5$
α	2
β	$2x^4 - 2x^3y + 2x^2y^2 - 2xy^3 + y^4 + y^3 - y - 1$
γ	$-y^4 + y^3 + y - 1$
$\{i, j\}$	$\{3, 17\}, k_{ij} = 2^4$
α	-1
β	$-x^4 + x^3y - x^2y^2 + xy^3 + y^4 + 4y^3 + 6y^2 + 11y + 20$
γ	$-y^4 - 2y^3 - 3y^2 - 5y - 10$
$\{i, j\}$	$\{3, 21\}, k_{ij} = 5$
α	1
β	$x^4 - x^3y + x^2y^2 - xy^2 - xy + x + y^3 + y^2 - 2y$
γ	$y^3 - y^2 - y + 1$
$\{i, j\}$	$\{3, 23\}, k_{ij} = 3$
α	$2x^5 + 4y + 3$
β	$2x^9 - 2x^8y + 2x^7y^2 + 2x^6y^2 + 2x^6y + 2x^6 - 2x^5y^3 - 2x^5y^2$ $- 4x^5y + 9x^4 - 9x^3y + 9x^2y^2 + 5xy^2 + 7xy + 9x - 5y^3 - 11y^2 - 12y$
γ	$2x^5y^3 + 2x^5y^2 + 2x^5y + 2x^5 + 9y^3 + 5y^2 + 7y + 9$
$\{i, j\}$	$\{3, 24\}, k_{ij} = 2^6 \cdot 3$
α	$5x^5 + 50y + 48$
β	$5x^9 - 5x^8y + 5x^7y^2 + 10x^6y^2 + 15x^6y + 25x^6 - 5x^5y^3 - 5x^5y^2$ $- 15x^5y + 25x^5 + 68x^4 - 68x^3y + 68x^2y^2 + 86xy^2 + 74xy + 100x - 18y^3$ $- 38y^2 - 74y + 100$
γ	$5x^5y^3 + 10x^5y^2 + 15x^5y + 25x^5 + 68y^3 + 86y^2 + 74y + 100$
$\{i, j\}$	$\{3, 33\}, k_{ij} = 2^3 \cdot 5$
α	$x^5 - y + 10$
β	$x^9 - x^8y + x^7y^2 + x^6y^2 - x^6 - 2x^5y^3 - x^5y^2 + x^5 + 14x^4$ $- 14x^3y + 14x^2y^2 + 15xy^2 + 6xy - 4x - 29y^3 - 20y^2 - 12y + 4$
γ	$x^5y^3 + x^5y^2 - x^5 + 14y^3 + 15y^2 + 6y - 4$
$\{i, j\}$	$\{3, 34\}, k_{ij} = 2^2 \cdot 5$
α	$-x + 5$
β	$-x^5 + x^4y + 5x^4 - x^3y^2 - 5x^3y + 6x^2y^2 + x^2 - 2xy^3 - 4xy^2 - 3x$ $+ 10y^3 - 5y^2 - 8$
γ	$-xy^3 + xy^2 + x + 5y^3 - 5y^2 - 4$
$\{i, j\}$	$\{4, 8\}, k_{ij} = 3$
α	1
β	$-x^4 - x + y^2 - y$
γ	$x^4 + 1$
$\{i, j\}$	$\{4, 14\}, k_{ij} = 1$
α	1
β	$-x^4 + x^3 - x^2 + y^2 - y - 1$
γ	$x^4 - x^3 + x^2 - x + 2$
$\{i, j\}$	$\{4, 16\}, k_{ij} = 5$
α	1
β	$x^4 + x^3 + x^2 + y^2 - y + 3$
γ	$-x^4 - x^3 - x^2 - x - 2$
$\{i, j\}$	$\{4, 22\}, k_{ij} = 2 \cdot 3^2$
α	$-x^4 + 5$
β	$x^7 - x^6 + 3x^5 - x^4y^2 + x^4y - 4x^4 + x^3 - x^2 - 3x + 5y^2 - 5y + 2$

$$\begin{aligned}
&\gamma \quad x^7 - x^6 + 2x^5 - 3x^4 + x^3 - x^2 + 2x - 3 \\
&\{i, j\} = \{4, 24\}, k_{ij} = 2 \\
&\alpha \quad 1 \\
&\beta \quad x^3 - x^2 - x + y^2 - y + 2 \\
&\gamma \quad -x^3 + x^2 - 1 \\
&\{i, j\} = \{4, 28\}, k_{ij} = 3 \\
&\alpha \quad 2x^4 + 1 \\
&\beta \quad -2x^7 + 2x^6 - 4x^5 + 2x^4y^2 - 2x^4y + 4x^4 - 5x^3 - x^2 + y^2 - y \\
&\gamma \quad -2x^7 + 2x^6 - 2x^5 + 2x^4 - 5x^3 - x^2 + x - 1 \\
&\{i, j\} = \{4, 33\}, k_{ij} = 3^2 \\
&\alpha \quad 1 \\
&\beta \quad 2x^3 + 2x^2 - 3x + y^2 - y - 5 \\
&\gamma \quad -x^3 - x^2 + x + 3 \\
&\{i, j\} = \{5, 8\}, k_{ij} = 5^2 \\
&\alpha \quad -1 \\
&\beta \quad x^4 - 2x^3 + 4x^2 - 7x - y^2 - y + 14 \\
&\gamma \quad -x^4 + 2x^3 - 4x^2 + 8x - 15 \\
&\{i, j\} = \{5, 9\}, k_{ij} = 5 \\
&\alpha \quad 1 \\
&\beta \quad -x^4 - x + y^2 + y + 2 \\
&\gamma \quad x^4 - 1 \\
&\{i, j\} = \{5, 16\}, k_{ij} = 5 \cdot 7^2 \\
&\alpha \quad 1 \\
&\beta \quad x^4 + 3x^3 + 9x^2 + 26x + y^2 + y + 81 \\
&\gamma \quad -x^4 - 3x^3 - 9x^2 - 27x - 80 \\
&\{i, j\} = \{5, 17\}, k_{ij} = 5 \cdot 7 \\
&\alpha \quad -32 \\
&\beta \quad 16x^4 - 24x^3 + 36x^2 - 22x - 32y^2 - 32y + 33 \\
&\gamma \quad -16x^4 + 24x^3 - 36x^2 + 54x - 65 \\
&\{i, j\} = \{5, 20\}, k_{ij} = 2 \cdot 7 \\
&\alpha \quad -x^4 + 2 \\
&\beta \quad x^7 + x^6 + x^5 - x^4y^2 - x^4y - 2x^4 - 4x^3 + 2x^2 + 4x + 2y^2 + 2y + 6 \\
&\gamma \quad x^7 + x^6 - x^4 - 4x^3 + 2x^2 + 6x + 4 \\
&\{i, j\} = \{5, 24\}, k_{ij} = 2 \cdot 3 \cdot 5 \\
&\alpha \quad -2x^4 + 7 \\
&\beta \quad -2x^7 + 2x^6 - 2x^5 - 2x^4y^2 - 2x^4y + 4x^4 - x^3 - 3x^2 - 5x + 7y^2 + 7y + 2 \\
&\gamma \quad 2x^7 - 2x^6 + 4x^5 - 6x^4 + x^3 + 3x^2 - 2x + 5 \\
&\{i, j\} = \{5, 26\}, k_{ij} = 5 \\
&\alpha \quad 1 \\
&\beta \quad -x^3 - 2x + y^2 + y + 1 \\
&\gamma \quad -x^3 - x \\
&\{i, j\} = \{5, 29\}, k_{ij} = 5 \\
&\alpha \quad 1 \\
&\beta \quad x^3 + x^2 + y^2 + y + 2 \\
&\gamma \quad -x^3 - x^2 - x - 1 \\
&\{i, j\} = \{5, 32\}, k_{ij} = 5^3 \\
&\alpha \quad -2x^4 + 27 \\
&\beta \quad -2x^7 + 2x^6 - 4x^5 - 2x^4y^2 - 2x^4y + 8x^4 + 7x^3 + 3x^2 - 16x + 27y^2 + 27y \\
&\quad + 22 \\
&\gamma \quad 2x^7 - 2x^6 + 6x^5 - 10x^4 - 7x^3 - 3x^2 - 11x + 5 \\
&\{i, j\} = \{5, 33\}, k_{ij} = 5^2 \cdot 7 \\
&\alpha \quad -2x^4 + 37 \\
&\beta \quad -4x^7 - 4x^6 - 10x^5 - 2x^4y^2 - 2x^4y - 22x^4 + 34x^3 - 26x^2 + 5x + 37y^2 \\
&\quad + 37y + 27 \\
&\gamma \quad 2x^7 + 2x^6 + 6x^5 + 10x^4 - 17x^3 + 13x^2 - 21x + 5 \\
&\{i, j\} = \{5, 34\}, k_{ij} = 3 \cdot 7 \\
&\alpha \quad 2x^4 - 1 \\
&\beta \quad -4x^7 + 4x^6 + 2x^5 + 2x^4y^2 + 2x^4y - 10x^4 + 10x^3 - 6x^2 - 13x - y^2 - y + 25 \\
&\gamma \quad -2x^7 + 2x^6 + 2x^5 - 6x^4 + 5x^3 - 3x^2 - 7x + 13
\end{aligned}$$

$$\begin{aligned}
&\{i, j\} = \{6, 15\}, k_{ij} = 2 \cdot 5 \\
&\alpha \quad 2 \\
&\beta \quad -2x^4 - 2x^3y - 2x^2y^2 - 2xy^3 - y^4 + y^3 + 2y^2 + 3y + 1 \\
&\gamma \quad y^4 + y^3 + 2y^2 + y + 1 \\
&\{i, j\} = \{6, 21\}, k_{ij} = 5 \\
&\alpha \quad 1 \\
&\beta \quad -x^4 - x^3y - x^2y^2 - xy^2 + xy - x - y^3 + y^2 \\
&\gamma \quad -y^3 + y^2 - y + 1 \\
&\{i, j\} = \{6, 23\}, k_{ij} = 5 \cdot 11 \\
&\alpha \quad 6x^5 - 36y - 5 \\
&\beta \quad -6x^9 - 6x^8y - 6x^7y^2 + 6x^6y^2 + 18x^6y + 18x^6 + 6x^5y^3 \\
&\quad + 18x^5y^2 + 24x^5y - 37x^4 - 37x^3y - 37x^2y^2 + xy^2 + 69xy + 75x + y^3 \\
&\quad + 33y^2 + 70y \\
&\gamma \quad -6x^5y^3 - 6x^5y^2 - 18x^5y - 18x^5 - 37y^3 - y^2 - 69y - 75 \\
&\{i, j\} = \{6, 27\}, k_{ij} = 2^3 \\
&\alpha \quad x + 3 \\
&\beta \quad -x^5 - x^4y - 3x^4 - x^3y^2 - 3x^3y - 4x^2y^2 + x^2 - xy^3 - 3xy^2 + 2xy + 2x \\
&\quad - 3y^3 + 6y - 2 \\
&\gamma \quad -xy^3 + xy^2 - x - 3y^3 + 3y^2 - 2 \\
&\{i, j\} = \{6, 30\}, k_{ij} = 2^4 \cdot 5 \\
&\alpha \quad 13x^5 - 169y - 76 \\
&\beta \quad -13x^9 - 13x^8y - 13x^7y^2 + 13x^6y^2 + 52x^6y + 65x^6 + 13x^5y^3 \\
&\quad + 52x^5y^2 + 78x^5y - 106x^4 - 106x^3y - 106x^2y^2 - 63xy^2 + 242xy + 192x \\
&\quad - 63y^3 + 73y^2 + 116y \\
&\gamma \quad -13x^5y^3 - 13x^5y^2 - 52x^5y - 65x^5 - 106y^3 + 63y^2 - 242y - 192 \\
&\{i, j\} = \{6, 34\}, k_{ij} = 2^4 \cdot 5^2 \cdot 11 \\
&\alpha \quad -31x^5 + 2883y + 1100 \\
&\beta \quad 31x^9 + 31x^8y + 31x^7y^2 + 93x^6y^2 - 310x^6y + 961x^6 + 62x^5y^3 \\
&\quad - 217x^5y^2 + 620x^5y + 961x^5 - 976x^4 - 976x^3y - 976x^2y^2 - 45xy^2 \\
&\quad - 1648xy + 3844x + 931y^3 + 1190y^2 + 3296y + 3844 \\
&\gamma \quad 31x^5y^3 - 93x^5y^2 + 310x^5y - 961x^5 - 976y^3 + 45y^2 + 1648y - 3844 \\
&\{i, j\} = \{7, 11\}, k_{ij} = 2^5 \cdot 13 \\
&\alpha \quad 32 \\
&\beta \quad 16x^4 - 8x^3y + 4x^2y^2 - 2xy^3 + y^4 - 16y \\
&\gamma \quad -y^4 - 2y^3 + 28y^2 + 72y + 144 \\
&\{i, j\} = \{7, 15\}, k_{ij} = 2^6 \cdot 11 \\
&\alpha \quad 16 \\
&\beta \quad 8x^4 - 4x^3y + 2x^2y^2 - xy^3 + y^4 + 2y^3 - 8y^2 - 48y - 160 \\
&\gamma \quad -y^4 - 4y^3 + 16y^2 + 80y + 320 \\
&\{i, j\} = \{7, 22\}, k_{ij} = 2^6 \cdot 3 \\
&\alpha \quad -40x^5 + 100y + 48 \\
&\beta \quad -20x^9 + 10x^8y - 5x^7y^2 + 5x^6y^2 + 70x^6y - 100x^6 - 5x^5y^3 \\
&\quad - 30x^5y^2 + 140x^5y - 100x^5 - 56x^4 + 28x^3y - 14x^2y^2 - 186xy^2 + 176xy \\
&\quad - 400x + 86y^3 - 324y^2 + 352y - 400 \\
&\gamma \quad -5x^5y^3 + 10x^5y^2 + 140x^5y - 200x^5 - 14y^3 - 372y^2 + 352y - 800 \\
&\{i, j\} = \{7, 25\}, k_{ij} = 2 \cdot 11 \\
&\alpha \quad -12x^5 + 15y^2 - 237y - 184 \\
&\beta \quad -6x^9 - 12x^8y + 6x^7y^2 - 18x^6y^2 + 24x^6 + 6x^5y^3 - 12x^5y^2 \\
&\quad - 21x^5y - 24x^5 - 242x^4 + 616x^3y - 308x^2y^2 + 1068xy^2 - 312xy + 229x + 15y^4 \\
&\quad - 485y^3 + 414y^2 + 717y - 379 \\
&\gamma \quad -30x^7y - 30x^6y + 6x^5y^3 - 6x^5y^2 + 48x^5 - 30x^3y^2 + 165x^3y \\
&\quad - 300x^3 - 30x^2y^2 + 1155x^2y - 300x^2 + 30xy^3 - 165xy^2 + 1290xy - 248y^3 \\
&\quad + 846y^2 - 189y + 758 \\
&\{i, j\} = \{7, 27\}, k_{ij} = 2^{10} \cdot 3 \\
&\alpha \quad 8x^5 - 64y \\
&\beta \quad 4x^9 - 2x^8y + x^7y^2 - 2x^6y^2 - 12x^6y + 56x^6 + x^5y^3 + 6x^5y^2 \\
&\quad - 32x^5y + 128x^4 - 64x^3y + 32x^2y^2 + 64xy^2 - 320xy + 768x - 32y^3 + 192y^2 \\
&\quad - 384y \\
&\gamma \quad x^5y^3 - 4x^5y^2 - 24x^5y + 112x^5 + 32y^3 + 128y^2 - 640y + 1536
\end{aligned}$$

$$\begin{aligned}
&\{i, j\} = \{8, 12\}, k_{ij} = 5 \cdot 13^2 \\
&\alpha \quad 243 \\
&\beta \quad 243x^4 - 243x^3y - 243x^3 + 243x^2y^2 + 486x^2y + 243x^2 - 243xy^3 - 729xy^2 \\
&\quad - 729xy - 243x + 81y^4 + 270y^3 + 468y^2 - 69y + 127 \\
&\gamma \quad -81y^4 - 351y^3 - 495y^2 - 399y - 58 \\
&\{i, j\} = \{8, 16\}, k_{ij} = 3^2 \cdot 5 \\
&\alpha \quad 32 \\
&\beta \quad 32x^4 - 32x^3y - 32x^3 + 32x^2y^2 + 64x^2y + 32x^2 - 32xy^3 - 96xy^2 - 96xy \\
&\quad - 32x + 16y^4 + 72y^3 + 132y^2 + 42y + 49 \\
&\gamma \quad -16y^4 - 56y^3 - 60y^2 - 54y + 17 \\
&\{i, j\} = \{8, 19\}, k_{ij} = 2 \cdot 3 \cdot 5 \\
&\alpha \quad x^5 - y + 8 \\
&\beta \quad x^9 - x^8y - x^8 + x^7y^2 + 2x^7y + x^7 - x^6y^2 - x^6y + x^5y^3 \\
&\quad + 2x^5y^2 + 12x^4 - 12x^3y - 12x^3 + 12x^2y^2 + 24x^2y + 12x^2 - 11xy^2 \\
&\quad - 10xy - 2x + 11y^3 + 22y^2 + 4y + 2 \\
&\gamma \quad x^5y^3 + 2x^5y^2 + 2x^5y + x^5 + 12y^3 + 25y^2 + 26y + 10 \\
&\{i, j\} = \{8, 26\}, k_{ij} = 2^3 \cdot 3^3 \\
&\alpha \quad -11x^5 - 121y + 318 \\
&\beta \quad -11x^9 + 11x^8y + 11x^8 - 11x^7y^2 - 22x^7y - 11x^7 + 11x^6y^2 - 11x^6y \\
&\quad + 22x^6 - 11x^5y^3 - 22x^5 + 274x^4 - 274x^3y - 274x^3 + 274x^2y^2 \\
&\quad + 548x^2y + 274x^2 - 153xy^2 - 12xy + 508x + 153y^3 + 286y^2 - 814y - 508 \\
&\gamma \quad -11x^5y^3 - 22x^5y^2 - 44x^5y + 11x^5 + 274y^3 + 669y^2 + 810y + 782 \\
&\{i, j\} = \{8, 29\}, k_{ij} = 3^4 \\
&\alpha \quad -4x + 7 \\
&\beta \quad -4x^5 + 4x^4y + 11x^4 - 4x^3y^2 - 15x^3y - 11x^3 + 7x^2y^2 + 26x^2y \\
&\quad + 11x^2 + 4xy^3 - 33xy + 5x - 7y^3 + 21y - 37 \\
&\gamma \quad -4xy^3 - 12xy^2 + 7y^3 + 21y^2 + 16 \\
&\{i, j\} = \{8, 33\}, k_{ij} = 1 \\
&\alpha \quad -1 \\
&\beta \quad -x^4 + x^3y + x^3 - x^2y^2 - 2x^2y - x^2 - xy^2 + 3x + 2y^3 + 5y^2 + y - 5 \\
&\gamma \quad -y^3 - 4y^2 - 3y + 2 \\
&\{i, j\} = \{8, 34\}, k_{ij} = 3^2 \cdot 5 \cdot 7 \\
&\alpha \quad -4x^5 - 16y + 101 \\
&\beta \quad -4x^9 + 4x^8y + 4x^8 - 4x^7y^2 - 8x^7y - 4x^7 + 4x^6y^2 + 4x^6 - 8x^5y^3 \\
&\quad - 12x^5y^2 - 12x^5y - 4x^5 + 89x^4 - 89x^3y - 89x^3 + 89x^2y^2 + 178x^2y \\
&\quad + 89x^2 - 73xy^2 - 44xy + 47x + 162y^3 + 327y^2 + 119y + 89 \\
&\gamma \quad -4x^5y^3 - 8x^5y^2 - 12x^5y + 89y^3 + 194y^2 + 223y + 136 \\
&\{i, j\} = \{9, 21\}, k_{ij} = 5 \\
&\alpha \quad 1 \\
&\beta \quad x^4 - x^3y + x^3 + x^2y^2 - 2x^2y + x^2 - xy^2 + x + y^3 - y^2 - 2y + 1 \\
&\gamma \quad y^3 - 4y^2 + 3y \\
&\{i, j\} = \{9, 23\}, k_{ij} = 5 \\
&\alpha \quad 1 \\
&\beta \quad x^4 - x^3y + x^3 + x^2y^2 - 2x^2y + x^2 + xy^2 - x - y^3 + y^2 - 1 \\
&\gamma \quad y^3 - 2y^2 + 3y - 2 \\
&\{i, j\} = \{10, 16\}, k_{ij} = 2^2 \\
&\alpha \quad -1 \\
&\beta \quad -x^4 + x - y^2 + 2y - 6 \\
&\gamma \quad x^4 + 2 \\
&\{i, j\} = \{10, 18\}, k_{ij} = 2^6 \\
&\alpha \quad 2 \\
&\beta \quad -x^4 - 2x^3 - 4x^2 - 10x + 2y^2 - 4y - 10 \\
&\gamma \quad x^4 + 2x^3 + 4x^2 + 8x + 18 \\
&\{i, j\} = \{11, 15\}, k_{ij} = 2^2 \cdot 3 \\
&\alpha \quad 1 \\
&\beta \quad -x^4 - x + y^2 + 2y + 6 \\
&\gamma \quad x^4 - 2 \\
&\{i, j\} = \{11, 17\}, k_{ij} = 2^5 \\
&\alpha \quad -2
\end{aligned}$$

$$\begin{aligned}
\beta & x^4 - 2x^3 + 4x^2 - 6x - 2y^2 - 4y + 6 \\
\gamma & -x^4 + 2x^3 - 4x^2 + 8x - 14 \\
\{i, j\} &= \{11, 22\}, k_{ij} = 2^3 \cdot 3^2 \\
\alpha & x^4 + 7 \\
\beta & -x^7 + x^6 + x^4y^2 + 2x^4y + x^4 - 4x^3 - 2x^2 + 3x + 7y^2 + 14y + 22 \\
\gamma & -x^7 + x^6 + x^5 - 3x^4 - 4x^3 - 2x^2 + 10x - 6 \\
\{i, j\} &= \{11, 23\}, k_{ij} = 3 \\
\alpha & 7x - 12 \\
\beta & 7x^4 - 12x^3 + 14x^2 + 7xy^2 + 14xy + 4x - 12y^2 - 24y + 1 \\
\gamma & -7x^4 + 12x^3 - 21x^2 + 36x - 49 \\
\{i, j\} &= \{12, 26\}, k_{ij} = 2^{11} \cdot 13 \\
\alpha & -24x^5 + 1152y + 2560 \\
\beta & 12x^9 + 6x^8y + 3x^7y^2 + 6x^6y^2 - 84x^6y + 408x^6 + 3x^5y^3 \\
& -42x^5y^2 + 192x^5y - 2048x^4 - 1024x^3y - 512x^2y^2 + 1280xy^2 + 896xy \\
& + 4096x + 640y^3 + 1024y^2 + 3328y \\
\gamma & 3x^5y^3 - 12x^5y^2 + 168x^5y - 816x^5 - 512y^3 - 2560y^2 - 1792y - 8192 \\
\{i, j\} &= \{12, 31\}, k_{ij} = 2^8 \cdot 3 \cdot 5 \\
\alpha & 8x^5 - 32y - 96 \\
\beta & -4x^9 - 2x^8y - x^7y^2 + 2x^6y^2 + 20x^6y + 56x^6 + x^5y^3 + 10x^5y^2 \\
& + 32x^5y + 144x^4 + 72x^3y + 36x^2y^2 - 136xy^2 - 560xy - 1056x - 68y^3 - 296y^2 \\
& - 576y \\
\gamma & -x^5y^3 - 4x^5y^2 - 40x^5y - 112x^5 + 36y^3 + 272y^2 + 1120y + 2112 \\
\{i, j\} &= \{13, 15\}, k_{ij} = 2^4 \cdot 3^2 \cdot 11 \\
\alpha & -16 \\
\beta & -8x^4 + 4x^3y + 4x^3 - 2x^2y^2 - 4x^2y - 2x^2 + xy^3 + 3xy^2 + 3xy + x - y^4 \\
& - 7y^3 - 17y^2 - 61y - 330 \\
\gamma & y^4 + 10y^3 + 28y^2 + 134y + 659 \\
\{i, j\} &= \{13, 16\}, k_{ij} = 17 \\
\alpha & -243 \\
\beta & -81y^4 - 675y^3 - 2196y^2 - 2739y - 1753 \\
\gamma & 243x^4 + 243x^3y + 486x^3 + 243x^2y^2 + 972x^2y + 972x^2 + 243xy^3 \\
& + 1458xy^2 + 2916xy + 1944x + 81y^4 + 594y^3 + 1440y^2 + 2055y + 382 \\
\{i, j\} &= \{13, 20\}, k_{ij} = 2^3 \cdot 17 \\
\alpha & -336x^5 + 522y^2 - 14130y - 5620 \\
\beta & -168x^9 - 438x^8y - 438x^7y^2 + 219x^7y^2 + 438x^7y + 219x^7 - 657x^6y^2 \\
& - 438x^6y + 1095x^6 + 219x^5y^3 - 168x^5y^2 - 573x^5y - 1230x^5 - 31055x^4 \\
& + 147898x^3y + 147898x^3 - 73949x^2y^2 - 147898x^2y - 73949x^2 + 232359xy^2 \\
& + 86578xy - 29728x + 522y^4 - 86513y^3 + 36055y^2 + 41056y - 22616 \\
\gamma & -1044x^7y - 1044x^7 - 1044x^6y - 1044x^6 + 219x^5y^3 + 387x^5y^2 - 219x^5y \\
& + 1365x^5 - 1044x^3y^2 + 10962x^3y - 56376x^3 - 1044x^2y^2 + 275703x^2y \\
& + 208365x^2 + 1044xy^3 - 12006xy^2 + 332079xy + 208365x - 71861y^3 - 44838y^2 \\
& + 75023y + 74960 \\
\{i, j\} &= \{13, 22\}, k_{ij} = 2^3 \cdot 5 \cdot 11 \\
\alpha & 3744x^5 - 6786y^2 - 17758y + 5016 \\
\beta & -1144x^9 - 182x^8y - 182x^8 + 91x^7y^2 + 182x^7y + 91x^7 - 91x^6y^2 \\
& - 1274x^6y + 4277x^6 + 91x^5y^3 + 5252x^5y^2 - 12805x^5y + 4446x^5 \\
& - 3421x^4 + 10976x^3y + 10976x^3 - 5488x^2y^2 - 10976x^2y - 5488x^2 \\
& + 24416xy^2 + 12040xy + 3817x - 6786y^4 + 10684y^3 - 12943y^2 - 47591y + 19624 \\
\gamma & -6032x^8 - 1508x^7y + 4524x^7 + 7540x^6y - 4524x^6 + 91x^5y^3 + 1599x^5y^2 \\
& - 14339x^5y + 13169x^5 + 13572x^3y^2 - 3770x^3y - 22388x^3 - 13572x^2y^2 \\
& + 22301x^2y + 40919x^2 - 13572xy^3 + 17342xy^2 + 87xy - 40919x + 21656y^3 - 7275y^2 \\
& - 33390y + 43065 \\
\{i, j\} &= \{13, 25\}, k_{ij} = 3^2 \cdot 5^2 \\
\alpha & -10348x^5 + 23205y^2 - 233441y - 256571 \\
\beta & -5174x^9 - 728x^8y - 728x^8 + 364x^7y^2 + 728x^7y + 364x^7 - 1092x^6y^2 \\
& - 1456x^6y + 9828x^6 + 364x^5y^3 - 4810x^5y^2 - 8047x^5y - 18239x^5 \\
& - 345327x^4 + 222516x^3y + 222516x^3 - 111258x^2y^2 - 222516x^2y - 111258x^2 \\
& + 409486xy^2 - 29624xy + 170521x + 23205y^4 - 275084y^3 + 67328y^2 + 279593y - 365775 \\
\gamma & -6630x^7y - 6630x^7 - 6630x^6y - 6630x^6 + 364x^5y^3 + 5538x^5y^2 \\
& + 4810x^5y + 20020x^5 - 46410x^3y^2 + 182325x^3y - 430950x^3 - 46410x^2y^2
\end{aligned}$$

$$\begin{aligned}
& +282030x^2y - 331245x^2 + 46410xy^3 - 182325xy^2 + 530655xy + 99705x - 18438y^3 \\
& +67253y^2 + 186848y + 660734 \\
\{i, j\} = \{13, 33\}, k_{ij} = 2^3 \cdot 5 \\
\alpha & 8 \\
\beta & 4x^4 - 2x^3y - 2x^3 + x^2y^2 + 2x^2y + x^2 + 3xy^2 + 4xy + 5x - 2y^3 - 8y^2 - 14y \\
& -8 \\
\gamma & y^3 + 9y^2 + 11y + 11 \\
\{i, j\} = \{14, 19\}, k_{ij} = 2^3 \cdot 3^2 \cdot 5 \\
\alpha & -7x^5 + 49y + 402 \\
\beta & -7x^9 + 7x^8y + 14x^8 - 7x^7y^2 - 28x^7y - 28x^7 + 7x^6y^2 + 14x^6y - 7x^6 \\
& -7x^5y^3 - 28x^5y^2 - 14x^5y + 14x^5 + 346x^4 - 346x^3y - 692x^3 \\
& +346x^2y^2 + 1384x^2y + 1384x^2 - 395xy^2 - 552xy - 312x + 395y^3 + 1293y^2 \\
& +1014y + 624 \\
\gamma & -7x^5y^3 - 35x^5y^2 - 70x^5y - 63x^5 + 346y^3 + 1681y^2 + 3600y + 2456 \\
\{i, j\} = \{14, 22\}, k_{ij} = 2 \cdot 3 \cdot 17 \\
\alpha & x^5 - 8y - 37 \\
\beta & x^9 - x^8y - 2x^8 + x^7y^2 + 4x^7y + 4x^7 + x^6y + 7x^6 + x^5y^3 \\
& +5x^5y^2 + 3x^5y + x^5 - 35x^4 + 35x^3y + 70x^3 - 35x^2y^2 - 140x^2y \\
& -140x^2 + 8xy^2 - 57xy - 155x - 43y^3 - 153y^2 - 171y - 125 \\
\gamma & x^5y^3 + 6x^5y^2 + 13x^5y + 15x^5 - 35y^3 - 202y^2 - 477y - 435 \\
\{i, j\} = \{14, 24\}, k_{ij} = 2^2 \cdot 5^2 \\
\alpha & 3x^5 - 18y - 22 \\
\beta & 3x^9 - 3x^8y - 6x^8 + 3x^7y^2 + 12x^7y + 12x^7 + 6x^6y^2 + 27x^6y + 39x^6 \\
& -3x^5y^3 - 15x^5y^2 - 33x^5y - 15x^5 + 20x^4 - 20x^3y - 40x^3 + 20x^2y^2 \\
& +80x^2y + 80x^2 + 58xy^2 + 204xy + 254x - 38y^3 - 124y^2 - 196y - 94 \\
\gamma & 3x^5y^3 + 24x^5y^2 + 63x^5y + 63x^5 + 20y^3 + 178y^2 + 444y + 414 \\
\{i, j\} = \{14, 33\}, k_{ij} = 2^3 \cdot 5 \cdot 17 \\
\alpha & -3x^5 - 9y - 140 \\
\beta & -3x^9 + 3x^8y + 6x^8 - 3x^7y^2 - 12x^7y - 12x^7 - 3x^6y^2 + 21x^6 \\
& +6x^5y^3 + 27x^5y^2 + 18x^5y - 39x^5 - 74x^4 + 74x^3y + 148x^3 \\
& -74x^2y^2 - 296x^2y - 296x^2 - 65xy^2 + 48xy + 404x + 139y^3 + 600y^2 + 480y - 620 \\
\gamma & -3x^5y^3 - 21x^5y^2 - 36x^5y - 3x^5 - 74y^3 - 509y^2 - 840y - 188 \\
\{i, j\} = \{15, 16\}, k_{ij} = 2^3 \cdot 3^2 \\
\alpha & 2 \\
\beta & 2x^4 - 2x^3y + 4x^3 + 2x^2y^2 - 8x^2y + 8x^2 - 2xy^3 + 12xy^2 - 24xy + 16x + y^4 \\
& -6y^3 + 9y^2 + 15y - 50 \\
\gamma & -y^4 + 10y^3 - 39y^2 + 81y - 82 \\
\{i, j\} = \{15, 22\}, k_{ij} = 2 \cdot 3^2 \cdot 11 \\
\alpha & x^5 - 8y + 13 \\
\beta & x^9 - x^8y + 2x^8 + x^7y^2 - 4x^7y + 4x^7 - 3x^6y - x^6 + x^5y^3 \\
& -3x^5y^2 + 3x^5y - 11x^5 - 5x^4 + 5x^3y - 10x^3 - 5x^2y^2 + 20x^2y - 20x^2 \\
& +8xy^2 + 37xy + 11x - 13y^3 + 25y^2 + 27y + 73 \\
\gamma & x^5y^3 - 6x^5y^2 + 9x^5y - 9x^5 - 5y^3 + 38y^2 - 23y + 51 \\
\{i, j\} = \{15, 25\}, k_{ij} = 3^2 \cdot 11 \\
\alpha & 4x^5 - 48y + 43 \\
\beta & 4x^9 - 4x^8y + 8x^8 + 4x^7y^2 - 16x^7y + 16x^7 - 8x^6y^2 + 32x^6y - 36x^6 \\
& +4x^5y^3 - 16x^5y^2 + 16x^5y - 4x^5 - 17x^4 + 17x^3y - 34x^3 - 17x^2y^2 \\
& +68x^2y - 68x^2 + 82xy^2 - 52xy + 21x - 65y^3 + 80y^2 + 88y - 115 \\
\gamma & 4x^5y^3 - 32x^5y^2 + 80x^5y - 68x^5 - 17y^3 + 184y^2 - 256y + 157 \\
\{i, j\} = \{15, 26\}, k_{ij} = 2^5 \cdot 3 \cdot 5 \\
\alpha & 7x^5 - 49y + 138 \\
\beta & 7x^9 - 7x^8y + 14x^8 + 7x^7y^2 - 28x^7y + 28x^7 - 7x^6y^2 + 28x^6y - 77x^6 \\
& +7x^5y^3 - 42x^5y^2 + 126x^5y - 154x^5 + 124x^4 - 124x^3y + 248x^3 \\
& +124x^2y^2 - 496x^2y + 496x^2 - 75xy^2 + 678xy - 1028x + 75y^3 - 779y^2 + 2246y \\
& -2056 \\
\gamma & 7x^5y^3 - 49x^5y^2 + 112x^5y - 133x^5 + 124y^3 - 819y^2 + 2166y - 2020 \\
\{i, j\} = \{15, 33\}, k_{ij} = 2^3 \cdot 13 \\
\alpha & -3x^5 - 9y + 40 \\
\beta & -3x^9 + 3x^8y - 6x^8 - 3x^7y^2 + 12x^7y - 12x^7 - 3x^6y^2 + 9x^6 \\
& +6x^5y^3 - 21x^5y^2 + 30x^5y - 15x^5 - 2x^4 + 2x^3y - 4x^3 - 2x^2y^2
\end{aligned}$$

$$\begin{aligned}
& +8x^2y - 8x^2 + 7xy^2 + 12xy - 12x - 5y^3 - 8y^2 + 8y - 28 \\
\gamma \quad & -3x^5y^3 + 15x^5y^2 - 36x^5y + 33x^5 - 2y^3 + 19y^2 - 12y + 4 \\
\{i, j\} = \{16, 19\}, k_{ij} = 2^5 \cdot 3 \cdot 5^2 \\
\alpha \quad & -7x^5 + 245y + 732 \\
\beta \quad & 7x^9 + 7x^8y + 14x^8 + 7x^7y^2 + 28x^7y + 28x^7 + 7x^6y^2 + 63x^6 \\
& + 7x^5y^3 + 14x^5y^2 + 56x^5y + 126x^5 - 452x^4 - 452x^3y - 904x^3 \\
& - 452x^2y^2 - 1808x^2y - 1808x^2 - 207xy^2 - 1190xy + 132x - 207y^3 - 1359y^2 \\
& - 1516y + 264 \\
\gamma \quad & 7x^5y^3 + 35x^5y^2 + 84x^5y - 7x^5 - 452y^3 - 2505y^2 - 4234y - 3748 \\
\{i, j\} = \{16, 20\}, k_{ij} = 2^4 \cdot 7 \cdot 17 \\
\alpha \quad & -5x^5 + 100y + 272 \\
\beta \quad & 5x^9 + 5x^8y + 10x^8 + 5x^7y^2 + 20x^7y + 20x^7 - 15x^6y + 5x^6 + 5x^5y^3 \\
& + 15x^5y^2 + 45x^5y + 45x^5 - 162x^4 - 162x^3y - 324x^3 - 162x^2y^2 \\
& - 648x^2y - 648x^2 + 100xy^2 + 96xy + 888x - 62y^3 - 676y^2 - 688y - 408 \\
\gamma \quad & 5x^5y^3 + 30x^5y^2 + 75x^5y + 35x^5 - 162y^3 - 1072y^2 - 2040y - 2184 \\
\{i, j\} = \{16, 30\}, k_{ij} = 2^5 \cdot 5 \\
\alpha \quad & -9x^5 + 81y + 122 \\
\beta \quad & 9x^9 + 9x^8y + 18x^8 + 9x^7y^2 + 36x^7y + 36x^7 - 9x^6y^2 - 54x^6y - 27x^6 \\
& - 9x^5y^3 - 72x^5y^2 - 144x^5y - 54x^5 + 4x^4 + 4x^3y + 8x^3 + 4x^2y^2 \\
& + 16x^2y + 16x^2 + 77xy^2 - 222xy + 132x + 77y^3 + 13y^2 - 190y + 264 \\
\gamma \quad & 9x^5y^3 + 63x^5y^2 + 162x^5y + 99x^5 + 4y^3 - 53y^2 + 270y - 100 \\
\{i, j\} = \{16, 34\}, k_{ij} = 2^6 \cdot 3^3 \cdot 5^2 \cdot 7 \\
\alpha \quad & -253x^5 + 64009y + 124742 \\
\beta \quad & 253x^9 + 253x^8y + 506x^8 + 253x^7y^2 + 1012x^7y + 1012x^7 + 759x^6y^2 \\
& - 506x^6y + 11385x^6 + 506x^5y^3 + 253x^5y^2 + 6578x^5y + 32131x^5 \\
& - 59468x^4 - 59468x^3y - 118936x^3 - 59468x^2y^2 - 237872x^2y - 237872x^2 \\
& - 114395xy^2 - 327862xy - 125820x - 54927y^3 - 250230y^2 - 271048y + 98284 \\
\gamma \quad & 253x^5y^3 + 759x^5y^2 + 3542x^5y - 9361x^5 - 59468y^3 - 242413y^2 - 385754y \\
& - 349924 \\
\{i, j\} = \{17, 22\}, k_{ij} = 2^8 \cdot 3 \cdot 7 \cdot 17 \\
\alpha \quad & -152x^5 + 4332y + 30312 \\
\beta \quad & -76x^9 + 38x^8y + 76x^8 - 19x^7y^2 - 76x^7y - 76x^7 + 19x^6y^2 + 266x^6y \\
& - 1444x^6 - 19x^5y^3 - 190x^5y^2 + 684x^5y - 76x^5 + 14928x^4 - 7464x^3y \\
& - 14928x^3 + 3732x^2y^2 + 14928x^2y + 14928x^2 - 12396xy^2 - 5508xy - 5928x \\
& + 8064y^3 + 11784y^2 + 10200y + 14928 \\
\gamma \quad & -19x^5y^3 - 76x^5y^2 + 304x^5y - 3040x^5 + 3732y^3 - 2400y^2 + 33768y + 18000 \\
\{i, j\} = \{17, 23\}, k_{ij} = 2^4 \cdot 3 \\
\alpha \quad & -8x^5 - 12y + 8 \\
\beta \quad & -4x^9 + 2x^8y + 4x^8 - x^7y^2 - 4x^7y - 4x^7 - 2x^6y^2 + 2x^6y - 28x^6 \\
& + x^5y^3 + x^5y^2 + 16x^5y + 28x^5 - 8x^4 + 4x^3y + 8x^3 - 2x^2y^2 \\
& - 8x^2y - 8x^2 + 20xy^2 - 32xy - 56x - 10y^3 + 2y^2 + 56y + 56 \\
\gamma \quad & -x^5y^3 - 10x^5y^2 - 8x^5y - 64x^5 - 2y^3 + 28y^2 - 88y - 128 \\
\{i, j\} = \{17, 30\}, k_{ij} = 2^9 \cdot 5 \\
\alpha \quad & -24x^5 - 72y - 784 \\
\beta \quad & -12x^9 + 6x^8y + 12x^8 - 3x^7y^2 - 12x^7y - 12x^7 - 6x^6y^2 - 144x^6 \\
& + 3x^5y^3 + 6x^5y^2 + 84x^5y + 144x^5 - 272x^4 + 136x^3y + 272x^3 \\
& - 68x^2y^2 - 272x^2y - 272x^2 + 8xy^2 + 552xy - 2304x - 4y^3 - 248y^2 + 992y + 2304 \\
\gamma \quad & -3x^5y^3 - 30x^5y^2 - 36x^5y - 312x^5 - 68y^3 - 392y^2 + 288y - 5152 \\
\{i, j\} = \{17, 32\}, k_{ij} = 2^5 \cdot 5^3 \\
\alpha \quad & 56x^5 + 980y + 3664 \\
\beta \quad & 28x^9 - 14x^8y - 28x^8 + 7x^7y^2 + 28x^7y + 28x^7 + 21x^6y^2 + 84x^6y \\
& + 616x^6 - 7x^5y^3 - 21x^5y^2 - 294x^5y + 28x^5 + 1272x^4 - 636x^3y \\
& - 1272x^3 + 318x^2y^2 + 1272x^2y + 1272x^2 - 1006xy^2 - 3324xy + 2784x + 662y^3 \\
& + 2126y^2 - 1316y + 1272 \\
\gamma \quad & 7x^5y^3 + 84x^5y^2 + 252x^5y + 1288x^5 + 318y^3 - 104y^2 - 2832y + 8112 \\
\{i, j\} = \{17, 33\}, k_{ij} = 2^7 \cdot 7 \\
\alpha \quad & -8x^5 + 28y + 248 \\
\beta \quad & -4x^9 + 2x^8y + 4x^8 - x^7y^2 - 4x^7y - 4x^7 - 3x^6y^2 - 6x^6y - 4x^6 \\
& + 2x^5y^3 + 12x^5y^2 + 24x^5y + 12x^5 + 96x^4 - 48x^3y - 96x^3 + 24x^2y^2 \\
& + 96x^2y + 96x^2 + 16xy^2 - 108xy - 72x - 20y^3 - 64y^2 - 16y + 48
\end{aligned}$$

$$\begin{aligned}
&\gamma \quad -x^5y^3 - 12x^5y^2 - 24x^5y - 16x^5 + 24y^3 + 176y^2 + 72y + 48 \\
&\{i, j\} = \{18, 23\}, k_{ij} = 2^2 \cdot 3^3 \cdot 5 \cdot 11 \\
&\alpha \quad 146y^2 - 1314y + 1188 \\
&\beta \quad 292x^6 - 146x^5y + 292x^5 - 458x^3 + 229x^2y - 458x^2 - 1328x + 146y^4 - 1022y^3 \\
&\quad - 1085y^2 + 17y - 734 \\
&\gamma \quad 584x^5 - 292x^3y^2 + 2482x^3y - 1188x^3 - 916x^2 - 292xy^3 + 2190xy^2 + 1294xy \\
&\quad - 1188x + 229y^2 - 1832y - 2656 \\
&\{i, j\} = \{18, 31\}, k_{ij} = 2^8 \cdot 3 \cdot 5^3 \cdot 11 \\
&\alpha \quad 376x^5 - 17672y + 123344 \\
&\beta \quad 188x^9 - 94x^8y + 188x^8 + 47x^7y^2 - 188x^7y + 188x^7 + 94x^6y^2 \\
&\quad - 752x^6y - 5264x^6 - 47x^5y^3 + 470x^5y^2 + 1692x^5y - 5264x^5 + 46256x^4 \\
&\quad - 23128x^3y + 46256x^3 + 11564x^2y^2 - 46256x^2y + 46256x^2 + 58472xy^2 \\
&\quad + 53736xy + 124608x - 29236y^3 + 40440y^2 - 70240y + 124608 \\
&\gamma \quad 47x^5y^3 - 94x^5y^2 - 940x^5y - 10904x^5 + 11564y^3 + 47560y^2 + 246240y \\
&\quad + 156704 \\
&\{i, j\} = \{18, 34\}, k_{ij} = 2^7 \cdot 3 \cdot 11 \\
&\alpha \quad 8x^5 + 4y - 392 \\
&\beta \quad 4x^9 - 2x^8y + 4x^8 + x^7y^2 - 4x^7y + 4x^7 - 3x^6y^2 + 6x^6y - 20x^6 \\
&\quad + 2x^5y^3 - 12x^5y^2 + 24x^5y - 44x^5 - 224x^4 + 112x^3y - 224x^3 \\
&\quad - 56x^2y^2 + 224x^2y - 224x^2 + 160xy^2 - 436xy + 1336x - 108y^3 + 704y^2 - 1680y \\
&\quad + 2896 \\
&\gamma \quad x^5y^3 - 12x^5y^2 + 24x^5y - 48x^5 - 56y^3 + 656y^2 - 1544y + 3120 \\
&\{i, j\} = \{19, 26\}, k_{ij} = 2 \\
&\alpha \quad 0 \\
&\beta \quad 1 \\
&\gamma \quad -1 \\
&\{i, j\} = \{19, 27\}, k_{ij} = 2 \\
&\alpha \quad 0 \\
&\beta \quad -1 \\
&\gamma \quad 1 \\
&\{i, j\} = \{19, 30\}, k_{ij} = 2 \cdot 5 \\
&\alpha \quad -4x + 6 \\
&\beta \quad 4x^4 - 6x^3 - 2x^2y - 4x^2 + 2xy^2 + xy + 8x - 3y^2 + 3y + 1 \\
&\gamma \quad -2x^2y + 4x^2 - 2xy^2 + 5xy - 8x + 3y^2 - 3y + 7 \\
&\{i, j\} = \{19, 31\}, k_{ij} = 2 \cdot 5 \\
&\alpha \quad 2 \\
&\beta \quad -2x^3 + xy - y^2 - y - 1 \\
&\gamma \quad xy + y^2 + y + 1 \\
&\{i, j\} = \{22, 24\}, k_{ij} = 2^3 \cdot 3 \\
&\alpha \quad 2x^4 + 6 \\
&\beta \quad -2x^7 + 2x^6 + x^5y + 2x^5 - x^4y^2 - 2x^4y + x^4 - 8x^3 + 4xy + 7x - 3y^2 - 4y \\
&\gamma \quad x^5y - 4x^5 + x^4y^2 - 2x^4y + x^4 + 4xy - 7x + 3y^2 - 4y \\
&\{i, j\} = \{22, 26\}, k_{ij} = 2^5 \cdot 5 \cdot 7 \\
&\alpha \quad -16y - 60 \\
&\beta \quad 2x^9 - 2x^8 - 2x^7 + 2x^6y + 2x^6 - 2x^5y^2 - 2x^5y + 23x^4 + 4x^3y^2 \\
&\quad + 22x^3y + 33x^3 - 4x^2y^2 - 22x^2y - 79x^2 - 4xy^3 - 18xy^2 - 11xy - 522x + 4y^4 \\
&\quad + 18y^3 - 5y^2 - 139y + 1202 \\
&\gamma \quad -2x^9 + 2x^5y^2 + 2x^5y - 2x^5 - 23x^4 + 601x - 4y^4 - 10y^3 + 45y^2 + 229y - 680 \\
&\{i, j\} = \{22, 30\}, k_{ij} = 2^6 \cdot 3 \cdot 5 \\
&\alpha \quad -656x^5 + 6888y^2 + 29356y + 28048 \\
&\beta \quad 82x^9 + 574x^8 - 164x^7y - 738x^7 - 410x^6y + 738x^6 + 410x^5y^2 - 1394x^5y \\
&\quad + 14247x^4 - 3444x^3y^2 - 25010x^3y - 41147x^3 + 3444x^2y^2 - 3484x^2y \\
&\quad + 12653x^2 + 3444xy^3 + 21566xy^2 + 44631xy + 30310x - 3444y^4 - 14678y^3 \\
&\quad - 15275y^2 - 45225y - 85926 \\
&\gamma \quad -82x^9 - 246x^5y^2 + 2542x^5y - 738x^5 - 14247x^4 - 42963x + 3444y^4 + 7790y^3 \\
&\quad - 2807y^2 - 12059y + 55616 \\
&\{i, j\} = \{22, 33\}, k_{ij} = 2^3 \cdot 5^2 \cdot 7 \cdot 17 \\
&\alpha \quad 6943356x^5 - 4654923y^2 + 1162477y + 5950 \\
&\beta \quad -1943342x^9 + 58528x^8 - 10974x^7y - 58528x^7 - 47554x^6y + 58528x^6 \\
&\quad + 5480864x^5y^2 - 2929645x^5y - 1586864x^5 - 19058603x^4 - 479198x^3y^2
\end{aligned}$$

$$\begin{aligned}
& -4493853x^3y + 42315558x^3 + 479198x^2y^2 - 10173549x^2y - 42315558x^2 \\
& + 479198xy^3 + 4014655xy^2 - 32142009xy + 519008x + 1551641y^4 + 1361897y^3 \\
& + 105144006y^2 + 19629488y + 4649908 \\
\gamma \quad & 1943342x^9 - 3115200x^8 + 5841000x^7y - 3115200x^7 - 389400x^6y - 3115200x^6 \\
& + 6212110x^5y^2 - 3584073x^5y - 1586864x^5 + 19058603x^4 + 895620x^3y^2 \\
& + 1744512x^3y - 31977726x^3 + 895620x^2y^2 + 73587723x^2y - 31977726x^2 \\
& + 1791240xy^3 + 4384644xy^2 + 9632271xy + 9818824x - 1551641y^4 + 1262187y^3 \\
& + 52574978y^2 + 504898y + 4649908 \\
\{i, j\} = \{24, 31\}, k_{ij} = 2^6 \cdot 3 \cdot 5^2 \\
\alpha \quad & 20 \\
\beta \quad & 2x^9 - 2x^8 + 6x^7 - 2x^6y - 6x^6 - 2x^5y^2 - 6x^5y + 51x^4 + 4x^3y^2 \\
& - 22x^3y - 83x^3 - 4x^2y^2 + 22x^2y + 185x^2 + 4xy^3 - 18xy^2 - 105xy - 910x \\
& + 4y^4 - 18y^3 - 105y^2 - 165y - 1450 \\
\gamma \quad & -2x^9 + 2x^5y^2 + 14x^5y + 6x^5 - 51x^4 + 725x - 4y^4 + 10y^3 + 145y^2 + 455y \\
& + 2360 \\
\{i, j\} = \{25, 27\}, k_{ij} = 3^3 \cdot 7 \\
\alpha \quad & -2y - 13 \\
\beta \quad & -x^9 - x^8 - 2x^7 + x^6y - x^6 + x^5y^2 - x^5 + 6x^4 - 2x^3y^2 - 5x^3y \\
& + 16x^3 - 2x^2y^2 - 5x^2y + 22x^2 + 2xy^3 + 5xy^2 - 16xy + 99x + 2y^4 + 7y^3 \\
& - 9y^2 + 3y - 87 \\
\gamma \quad & x^9 - x^5y^2 + 3x^5y + 2x^5 - 6x^4 - 93x - 2y^4 - 3y^3 + 21y^2 - 36y + 164 \\
\{i, j\} = \{25, 32\}, k_{ij} = 2 \cdot 3^2 \\
\alpha \quad & -8x^5 - 16y^2 - 40y - 18 \\
\beta \quad & -4x^9 + 8x^7y + 8x^7 + 8x^6y + 8x^6 + 8x^5y + 4x^5 - x^4 - 16x^3y^2 + 8x^3y \\
& + 40x^3 - 16x^2y^2 + 10x^2y + 42x^2 + 16xy^3 - 8xy^2 - 38xy + 46x + 8y^4 + 28y^3 \\
& - 4y^2 - 10y + 69 \\
\gamma \quad & 4x^9 - 8x^5y^2 - 16x^5y - 4x^5 + x^4 - 44x - 8y^4 + 4y^3 + 2y^2 - 80y - 21 \\
\{i, j\} = \{26, 30\}, k_{ij} = 2^3 \\
\alpha \quad & 2x - 2 \\
\beta \quad & -2x^4 + 2x^3 + x^2y + x^2 - xy^2 - xy - x + y^2 - 4 \\
\gamma \quad & x^2y - 5x^2 + xy^2 - xy + 5x - y^2 - 4 \\
\{i, j\} = \{26, 33\}, k_{ij} = 2^2 \cdot 5 \cdot 17^2 \\
\alpha \quad & 9565776x^5 + 100845810y^2 - 59129313y - 32149565 \\
\beta \quad & -2922876x^9 - 23668x^6 + 8502912x^5y^2 - 44835000x^5y - 12512320x^5 \\
& - 20651517x^4 + 106506x^3y^2 + 1094645x^3y + 2750144x^3 - 4116777x^2y \\
& + 1372259x^2 - 106506xy^3 - 881633xy^2 + 811405xy - 16634727x - 33615270y^4 \\
& - 25181593y^3 + 28405879y^2 - 183679090y - 64302020 \\
\gamma \quad & 2922876x^9 - 6642900x^8 + 8768628x^7y - 12488652x^7 - 4517172x^6y \\
& - 12464984x^6 + 9034344x^5y^2 + 15340524x^5y + 47336x^5 + 20651517x^4 \\
& - 8430444x^3y^2 - 19028340x^3y - 13236806x^3 - 8430444x^2y^2 + 47042988x^2y \\
& - 55912099x^2 - 16860888xy^3 - 46487124xy^2 + 19197117xy - 33777084x + 33615270y^4 \\
& - 8540183y^3 + 23309750y^2 + 81446196y + 33269454 \\
\{i, j\} = \{27, 31\}, k_{ij} = 2^3 \cdot 3 \\
\alpha \quad & -2x + 2 \\
\beta \quad & 2x^4 - 2x^3 - x^2y - x^2 + xy^2 + xy + x - y^2 + 4 \\
\gamma \quad & -x^2y - 3x^2 - xy^2 + xy + 3x + y^2 + 4 \\
\{i, j\} = \{28, 31\}, k_{ij} = 3^6 \\
\alpha \quad & -8x^5 + 36y^2 + 44y + 1073 \\
\beta \quad & x^9 + 7x^8 - 2x^7y - 6x^7 - 5x^6y - x^6 + 5x^5y^2 - 2x^5y + x^5 - 119x^4 \\
& - 18x^3y^2 - 67x^3y - 935x^3 + 18x^2y^2 + 305x^2y + 816x^2 + 18xy^3 \\
& + 67xy^2 + 697xy + 723x - 18y^4 - 49y^3 - 720y^2 - 951y + 485 \\
\gamma \quad & -x^9 - 3x^5y^2 + 15x^5y + 6x^5 + 119x^4 - 604x + 18y^4 + 13y^3 + 330y^2 - 867y \\
& - 2024 \\
\{i, j\} = \{29, 33\}, k_{ij} = 3^2 \\
\alpha \quad & 1 \\
\beta \quad & -x^3 - x^2 - 2xy + x - y^2 - 3y + 5 \\
\gamma \quad & xy - x + y^2 + y - 4 \\
\{i, j\} = \{30, 32\}, k_{ij} = 1 \\
\alpha \quad & 1 \\
\beta \quad & -x^6 - x^3y - 2x^3 - xy^2 - 10xy - 9x + y^3 + 6y^2 + 11y + 8
\end{aligned}$$

γ	$x^6 - x^5 + 6xy + 5x - y^3 - 6y^2 - 17y - 13$
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Table 3: The polynomials $\varphi_i(x) = k_i^3 F(x, -r_i(x)/k_i)$ of Step 2 and the signs of H_i for $x \geq 5$ and for $x \leq -5$.

i	k_i	$\varphi_i(x)$	$x \geq 5$	$x \leq -5$
0	-1	$-x^6 + 2x^5 - 3x^4 + 2x^3 - x^2 - 4$	+	+
1	-1	$-x^6 - 4x^5 - 6x^4 - 6x^3 - 5x^2 - 2x - 5$	+	+
2	-1	$-x^5 - 4$	+	-
3	-1	$-x^5 + x^3 - x^2 - 4$	+	-
4	-1	$-x^5 - x - 3$	+	-
5	-1	$-x^5 + x - 5$	+	-
6	-1	$-x^5 - x^3 + x^2 - 4$	+	-
7	-1	$-x^5 + 8x^3 - 2x^2 - 4$	+	-
8	-1	$-x^5 + x^3 + 2x^2 + 2x - 3$	+	-
9	-1	$-x^5 + x^3 - 4x^2 + 4x - 5$	+	-
10	-1	$-x^5 - 2x + 4$	+	-
11	-1	$-x^5 + 2x - 12$	+	-
12	-1	$-x^5 - 8x^3 + 2x^2 - 4$	+	-
13	-1	$-x^5 + 8x^3 + 10x^2 + 5x - 3$	+	-
14	-1	$-x^5 + x^3 + 5x^2 + 10x + 4$	+	-
15	-1	$-x^5 + x^3 - 7x^2 + 14x - 12$	+	-
16	-1	$-x^5 - x^3 + 7x^2 - 14x + 4$	+	-
17	-1	$-x^5 + 8x^3 + 22x^2 + 22x + 4$	+	-
18	-1	$-x^5 + 8x^3 - 26x^2 + 26x - 12$	+	-
19	1	$-x^6 + x^5 + x^3 + 4$	+	+
20	1	$-x^6 + 4x^5 - 3x^4 + 2x^3 - x^2 + 4$	+	+
21	1	$-x^6 + x^5 - 3x^4 + x^3 - 3x^2 + x + 3$	+	+
22	1	$-x^6 - 2x^5 - 3x^4 + x^2 + 4$	+	+
23	-1	$-x^6 - x^5 + 3x^4 + x^3 - 3x^2 - x - 3$	+	+
24	-1	$-x^6 - 4x^5 - 3x^4 + x^2 - 4$	+	+
25	1	$-x^6 + 4x^5 - 6x^4 + 8x^3 - 7x^2 + 4x + 3$	+	+
26	1	$-x^6 + x^5 + 6x^4 + x^3 - 12x^2 - 2x + 12$	+	+
27	1	$-x^6 + x^5 - 6x^4 + x^3 - 12x^2 + 2x - 4$	+	+
28	1	$-x^6 - 2x^5 - 6x^4 - 6x^3 - 5x^2 - 2x + 3$	+	+
29	-1	$-x^6 + 2x^5 - 6x^4 + 8x^3 - 7x^2 + 4x - 5$	+	+
30	-1	$-x^6 - x^5 + 6x^4 + x^3 - 12x^2 - 2x + 4$	+	+
31	-1	$-x^6 - x^5 - 6x^4 + x^3 - 12x^2 + 2x - 12$	+	+
32	-1	$-x^6 - 4x^5 + 6x^3 + x^2 - 4x - 3$	+	+
33	-2	$-x^6 - 5x^5 - 3x^4 + 5x^3 - 4x^2 - 32$	+	+
34	2	$-x^6 + 5x^5 - 3x^4 + 3x^3 + 4x^2 + 32$	+	+

Table 4: The terminal residue classes $x \equiv a, y \equiv b \pmod{p^k}$ of Step 4, grouped by p, k and the value of B_p .

p	k	B_p	classes (a, b)
2	6	-1	(10, 6), (10, 38), (14, 42), (26, 22), (30, 26), (42, 6), (46, 10), (58, 54), (62, 58)
2	7	-1	(2, 46), (2, 110), (6, 50), (6, 114), (22, 34), (22, 98), (34, 14), (34, 78), (38, 18), (38, 82), (54, 2), (54, 66), (66, 46), (66, 110), (70, 50), (70, 114), (86, 34), (86, 98), (98, 14), (98, 78), (102, 18), (102, 82), (118, 2), (118, 66)
2	8	-1	(18, 94), (18, 222), (50, 62), (50, 190), (82, 30), (82, 158), (114, 126), (114, 254), (146, 94), (146, 222), (178, 62), (178, 190), (210, 30), (210, 158), (242, 126), (242, 254)
3	3	1	(2, 18), (6, 2), (11, 0), (14, 3), (15, 5), (15, 14), (15, 23), (20, 9), (23, 12), (24, 17), (24, 26)

p	k	B_p	classes (a, b)
3	4	1	(5, 48), (8, 33), (17, 42), (24, 8), (24, 35), (24, 62), (26, 51), (32, 75), (33, 11), (33, 38), (33, 65), (35, 60), (44, 69), (53, 78), (60, 20), (60, 47), (60, 74), (62, 6), (71, 15), (80, 24)
3	5	1	(59, 183), (140, 21), (221, 102)
5	2	1	(0, 16), (1, 5), (1, 9), (1, 11), (2, 16), (4, 1), (5, 1), (6, 1), (6, 5), (6, 19), (7, 21), (9, 21), (10, 11), (11, 4), (11, 5), (11, 16), (12, 1), (13, 13), (14, 16), (15, 21), (16, 5), (16, 6), (18, 23), (19, 11), (20, 6), (21, 5), (21, 21), (21, 24), (22, 2), (22, 11), (22, 12), (22, 17), (23, 8), (23, 11), (23, 16), (24, 6)
5	3	1	(3, 43), (8, 28), (16, 89), (17, 31), (23, 21), (23, 46), (23, 71), (23, 96), (23, 121), (28, 93), (33, 78), (41, 14), (42, 56), (47, 22), (47, 47), (47, 72), (47, 97), (47, 122), (53, 18), (58, 3), (66, 64), (67, 81), (73, 6), (73, 31), (73, 56), (73, 81), (73, 106), (78, 68), (83, 53), (91, 114), (92, 106), (98, 1), (98, 26), (98, 51), (98, 76), (98, 101), (103, 118), (108, 103), (116, 39), (117, 6), (122, 32), (122, 57), (122, 82), (122, 107)
5	4	1	(372, 7), (372, 132), (372, 257), (372, 382), (372, 507)
7	1	1	(3, 5), (4, 3), (5, 0), (6, 5)
7	2	1	(2, 15), (3, 8), (3, 15), (3, 22), (3, 29), (3, 36), (3, 43), (9, 1), (16, 36), (23, 22), (30, 8), (37, 43), (44, 29)
7	3	1	(297, 1), (297, 50), (297, 99), (297, 148), (297, 197), (297, 246), (297, 295)
11	1	1	(0, 6), (1, 2), (1, 7), (4, 5), (7, 7), (8, 9), (10, 3)
11	2	1	(8, 30), (8, 60), (9, 59), (19, 16), (19, 74), (20, 81), (30, 93), (30, 118), (31, 103), (41, 41), (41, 49), (42, 4), (52, 5), (52, 85), (53, 26), (63, 8), (63, 82), (64, 48), (74, 38), (74, 52), (75, 70), (85, 96), (85, 115), (86, 92), (96, 19), (96, 71), (97, 114), (107, 27), (107, 63), (108, 15), (118, 104), (118, 107), (119, 37)
13	1	1	(1, 4), (3, 0), (3, 4), (3, 9), (5, 6), (5, 8), (5, 12), (8, 7), (9, 6), (12, 9)
13	2	1	(4, 151), (17, 99), (30, 47), (43, 164), (51, 2), (51, 15), (51, 28), (51, 41), (51, 54), (51, 67), (51, 80), (51, 93), (51, 106), (51, 119), (51, 132), (51, 145), (51, 158), (69, 60), (82, 8), (95, 125), (108, 73), (121, 21), (134, 138), (147, 86), (160, 34)
13	3	1	(56, 450), (225, 1971), (394, 1295), (563, 619), (732, 2140), (901, 1464), (1070, 788), (1239, 112), (1408, 1633), (1577, 957), (1746, 281), (1915, 1802), (2084, 1126)
17	1	1	(0, 4), (3, 5), (4, 7), (5, 6), (9, 12), (10, 4), (10, 5), (10, 8), (11, 2), (11, 11), (13, 0), (13, 8), (13, 9)
17	2	1	(6, 38), (7, 157), (11, 259), (23, 4), (24, 242), (28, 72), (40, 259), (41, 38), (45, 174), (57, 225), (58, 123), (62, 276), (74, 191), (75, 208), (79, 89), (91, 157), (92, 4), (96, 191), (108, 123), (109, 89), (113, 4), (125, 89), (126, 174), (130, 106), (142, 55), (143, 259), (147, 208), (159, 21), (160, 55), (164, 21), (176, 276), (177, 140), (181, 123), (193, 242), (198, 225), (210, 208), (211, 21), (215, 38), (227, 174), (228, 106), (232, 140), (244, 140), (245, 191), (249, 242), (261, 106), (262, 276), (266, 55), (278, 72), (279, 72), (283, 157)
17	3	1	(194, 3404), (483, 4849), (772, 1381), (1061, 2826), (1350, 4271), (1639, 803), (1928, 2248), (2217, 3693), (2506, 225), (2795, 1670), (3084, 3115), (3373, 4560), (3662, 1092), (3951, 2537), (4240, 3982), (4529, 514), (4818, 1959)